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COMPARATIVE IN-SILICO ANALYSIS OF TP53 MUTATION HOTSPOTS IN HUMANS AND ELEPHANTS

A Thesis Submitted
In partial fulfilment of the requirements
for the award of the degree of
MASTER OF SCIENCE (BIOINFORMATICS)

Under the guidance of
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Mumbai, Maharashtra, India

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DEPARTMENT OF BIOINFORMATICS

CERTIFICATE

This is to certify that the thesis entitled **"Comparative In-Silico Analysis of TP53 Mutation Hotspots in Humans and Elephants"**, submitted by **Miss Ritika Rajendra Rawat** (Roll No: **B-131**), in partial fulfillment of the requirements for the award of the degree of **M.Sc. Bioinformatics (Part II – Semester IV)** of the **University of Mumbai**, is a record of original research work carried out by her under my supervision and guidance during the academic year **2025–2026**.

Mentor's Name (Mrs. Sermarani Nadar)	**Head of Department** (Dr. Gursimran Kaur Uppal)	**External Examiner**

DECLARATION

I hereby declare that the thesis entitled **"Comparative In-Silico Analysis of TP53 Mutation Hotspots in Humans and Elephants"** submitted to the **University of Mumbai** in partial fulfilment of the requirements for the award of the degree of **Master of Science (Bioinformatics)** is a record of original research work carried out by me under the academic guidance of **Mrs. Sermarani Nadar (Mentor)** and **Dr. Gursimran Kaur Uppal (Head of Department)**, under the academic administration of **Dr. Ratna Sharma, Principal, Guru Nanak Khalsa College, Mumbai**.

I further declare that this work has not been previously submitted to this or any other University or Institution for the award of any degree, diploma, or other academic qualification. The information derived from published or unpublished sources has been duly acknowledged and cited in the text and references.

I take full responsibility for the authenticity and accuracy of the data presented in this thesis.

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LIST OF ABBREVIATIONS

TP53 – Tumor Protein 53
p53 – Tumor Suppressor Protein 53
DNA – Deoxyribonucleic Acid
RNA – Ribonucleic Acid
MSA – Multiple Sequence Alignment
NCBI – National Center for Biotechnology Information
COSMIC – Catalogue of Somatic Mutations in Cancer
TCGA – The Cancer Genome Atlas
IARC – International Agency for Research on Cancer
DBD – DNA-Binding Domain
TAD – Transactivation Domain
RTG – Retrogene
ORF – Open Reading Frame
FASTA – Text-Based Sequence Format
CSV – Comma-Separated Values

ABSTRACT

Cancer doesn't just follow a simple rule where bigger animals or those that live longer get more of it. Peto's paradox flips that idea on its head. Take elephants—they're huge, they live for decades, but cancer rarely takes them down. Scientists have linked part of this resistance to elephants carrying extra copies of the TP53 gene.

In our study, we dug into this by focusing on well-known human TP53 hotspot codons—R175, G245, R248, R249, R273, and R282—and checked how these sites line up with both canonical and retrogene TP53 sequences from African and Asian elephants. We ran multiple sequence alignments at the protein level, then mapped out the codons to see which hotspot residues matched up and which didn't. After sorting all this data into clear datasets, we built a scoring system to highlight each hotspot's importance, combining how often mutations show up with how well those spots are conserved.

What did we find? The elephants' main TP53 genes keep essential DNA-binding domain residues nearly unchanged, while their retrogene copies show more differences. This work lays out a reproducible computational method and detailed comparative tables, setting the stage for future studies on TP53 function and structure.

KEYWORDS

TP53 tumour suppressor gene; comparative genomics; cancer mutation hotspots; elephant TP53 retrogenes; Peto's paradox; in-silico bioinformatics analysis; evolutionary cancer biology.

CHAPTER 1
INTRODUCTION

1.1 Cancer and Tumor Suppressor Genes

Cancer isn't a single disease—it's a complex breakdown in how cells behave. Instead of following the usual rules—grow, divide, then die when it's time—cancer cells ignore every stop sign. They multiply endlessly, slip past the normal self-destruct triggers, and rack up genetic damage. These cells don't stay in their lane, either. They invade nearby tissues and can spread far from where they started.

The body usually keeps cell growth under strict control. Molecular signals tell cells when to divide, when to stop, and when to destroy themselves if something's off. These systems also scan for DNA mistakes, fix what they can, or order a cell to die if the damage can't be repaired. Thanks to this constant surveillance, abnormal cells rarely get a chance to cause problems.

Cancer upends all of that. Over time, cells build up a mix of genetic and epigenetic changes—mutations, insertions, deletions, chromosome rearrangements, copy number changes, or shifts in gene expression. As this damage piles up, cells lose their normal checks and start acting out, surviving and growing in conditions that would usually wipe them out.

Two main types of genes run this show: oncogenes and tumour suppressor genes. Proto-oncogenes, which help regulate healthy cell growth, can turn into oncogenes if they mutate or get overactive. When that happens, they drive relentless growth and help cells avoid death. Tumor suppressor genes work in the opposite direction. They hold cell division in check, guard the genome, pause the cell cycle, repair DNA, and trigger cell death when things go wrong.

Tumor suppressors are essential. Lose them, and it's like removing the brakes from a speeding car—damaged cells just keep dividing and picking up more mutations. The "two-hit hypothesis" explains that both copies of a tumor suppressor gene usually need to be lost before cancer develops. That's how critical these genes are for keeping cell growth under control.

Among tumor suppressors, TP53 is the heavyweight. People call it the "guardian of the genome" for a reason. TP53 leads the response when DNA gets damaged or when there are signs of trouble. It pauses the cell cycle, calls for repairs, or orders cell death if the damage is beyond repair. Because TP53 plays such a central role in cancer defence, researchers have studied it more than almost any other gene in oncology.

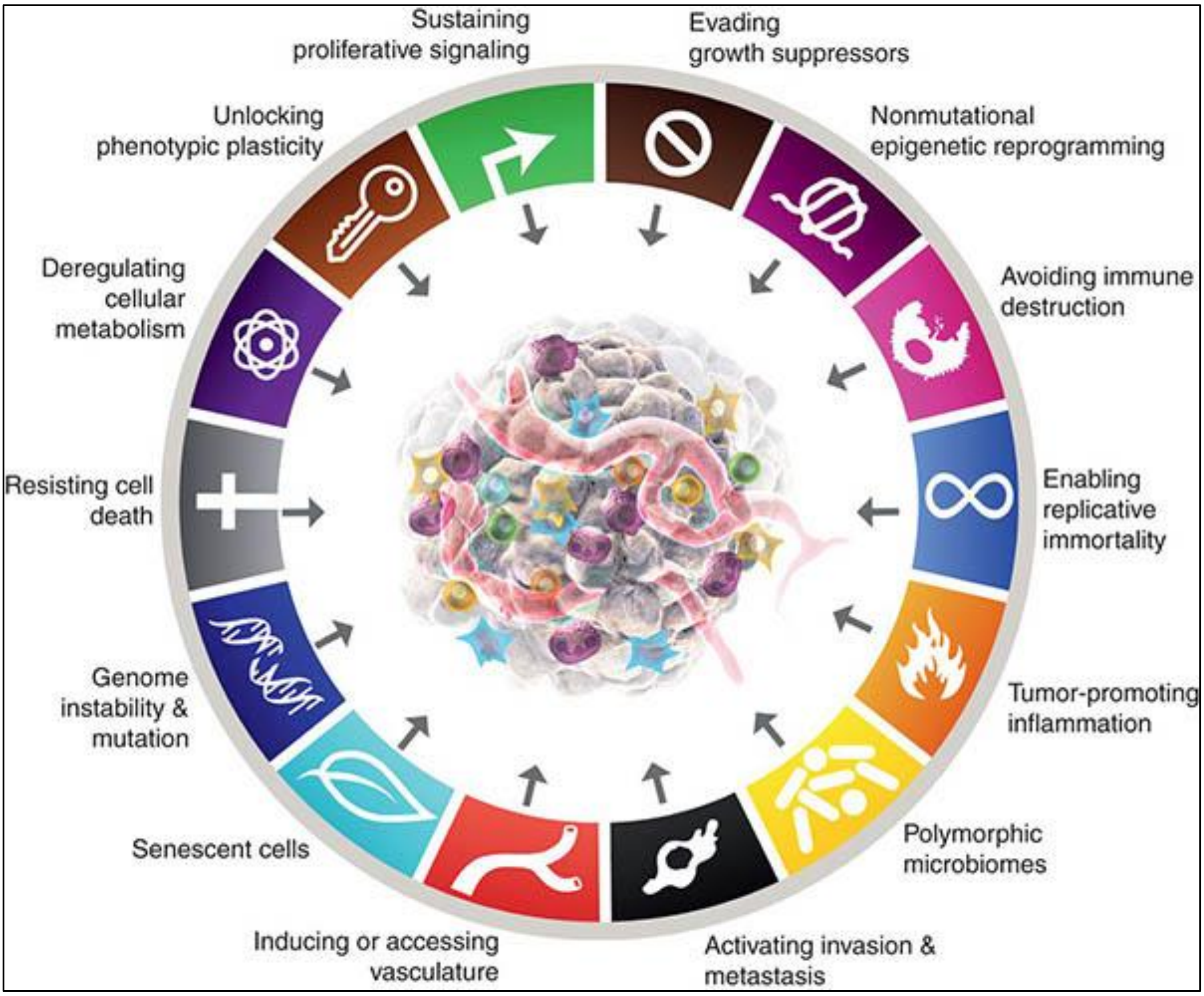

Figure 1.1: Hallmarks of cancer illustrating key biological processes involved in tumor progression ([1]).

1.2 TP53 Gene: Structure and Biological Function

The TP53 gene produces the p53 protein, a transcription factor so crucial it has been dubbed the "guardian of the genome." That title is well deserved—p53 truly does safeguard our DNA. When cells face problems—such as DNA damage, oxidative stress, low oxygen, ribosomal disruptions, or overactive oncogenes—p53 springs into action to help maintain order. Most of the time, however, cells don't allow p53 to linger. They keep its levels low, primarily by employing the E3 ubiquitin ligase MDM2 to mark p53 for destruction in the proteasome. This continuous turnover prevents p53 from inhibiting cell growth when there is no actual danger.

When things go wrong—such as when DNA is damaged—p53 springs into action. It stabilizes rapidly. Phosphorylation and acetylation begin to operate, and soon, p53 accumulates in the nucleus.

When p53 appears and starts functioning, it attaches to specific regions on the DNA and takes command of a range of target genes. These genes make crucial decisions for the cell—stopping the cell cycle, repairing DNA damage, initiating apoptosis, driving the cell into senescence, or adjusting metabolism. For example, if p53 activates a CDK inhibitor like p21, the cell cycle pauses at the G1/S or G2/M checkpoint, giving the cell time to repair its DNA. But if the damage is too extensive, p53 does not hesitate. It activates genes like BAX and PUMA, steering the cell toward apoptosis. This ability to either pause the process or trigger cell death is central to p53's function as a tumor suppressor.

Examining p53's structure reveals it is designed for this role. The N-terminal transactivation domain recruits co-activators and other proteins that assist in gene activation. The central DNA-binding domain serves as the main functional part—it locates and binds the correct DNA sequences, which is absolutely essential. Additionally, the C-terminal tetramerization domain allows four p53 molecules to assemble into a tetramer, enabling much tighter DNA binding. Other regions in the C-terminus fine-tune how p53 binds DNA and help maintain protein stability.

Most TP53 mutations associated with cancer affect the DNA-binding domain. When this region is impaired, p53 cannot bind DNA as it should. It loses its ability to regulate the genes that control the cell cycle or trigger cell death. Without this regulation, cells begin to grow uncontrollably, leading to tumor development and progression.

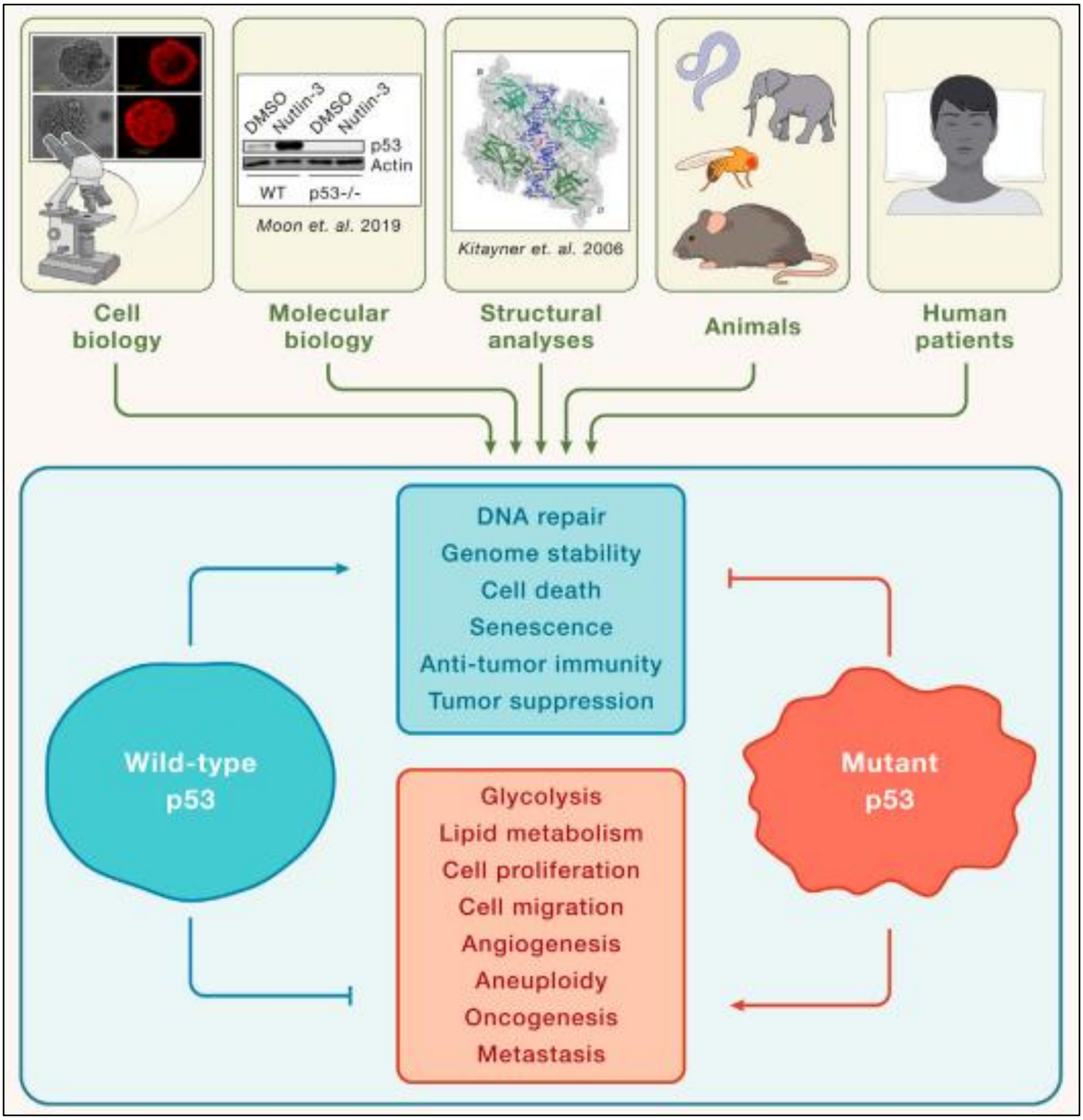

Figure 1.2: Structural organization of the TP53 protein showing major functional domains ([2]).

1.3 TP53 Mutation Hotspots in Human Cancer

TP53 is one of the most frequently mutated genes in human cancer. What's striking is that, instead of cropping up all over the place, these mutations tend to cluster at a handful of specific codons—mutation hotspots. You keep seeing the same hotspots in all sorts of cancers, which shows there's heavy selective pressure as tumors form and evolve.

The big six hotspots everyone talks about are codons R175, G245, R248, R249, R273, and R282. Most of them fall in the DNA-binding domain—right where the protein needs to hold its shape or latch onto DNA. When these codons change, it's usually an amino acid substitution. That single switch can wreck the protein's ability to bind DNA or stay stable, wiping out its tumor suppressor job.

Look at R248. Mutations at this spot hit the part of the protein that grabs DNA directly. Over at R175, changes disrupt the core structure and throw everything off. Huge cancer genomics projects like COSMIC and TCGA keep finding these same hotspots in different cancers, again and again.

Because these recurring mutations have such a big impact on TP53's function, researchers often focus on them in comparative genomic studies across different species.

Mutation type	Expected (count)	Expected (%)	Observed (count)	Observed (%)
Missense	1,150	73.4	17,191	87.9
Nonsense	58	3.7	1,435	7.3
Silent	359	22.9	932	4.8
Total	**1,567**	**100**	**19,558**	**100**

Figure 1.3: Frequently reported TP53 mutation hotspot codons localized within the DNA-binding domain (Adapted from [3]).

1.4 Peto's Paradox and Cancer Resistance in Large Mammals

Cancer doesn't behave according to the numbers we might expect. Looking across various species, larger animals with longer lifespans—like elephants and Cancer doesn't follow the script we'd predict. You'd think that big animals—elephants, whales, creatures with far more cells and longer lives—would be swimming in cancer cases compared to smaller species. More cells, more time, more chances for things to go haywire. Yet that's not what happens. Elephants aren't dropping dead from tumors. Whales aren't riddled with cancer, either. This weird mismatch between expectation and reality is what scientists call Peto's paradox.

So, what's really happening? It turns out, large animals have evolved powerful ways to keep cancer in check. Evolution pushes them to build better tumor suppression systems—without those, their sheer size and long lifespan would make them sitting ducks for cancer. Over countless generations, these animals have upgraded their genome-protection strategies, making sure cell growth doesn't spiral out of control.

Look at elephants. They can live more than sixty years and their bodies are enormous, but their cancer rates stay impressively low. The key is in their DNA. Elephants have many copies of the TP53 gene—far more than humans. These extra copies come from ancient gene duplications, and they give elephants a turbocharged response to DNA damage. In the lab, when elephant cells sense DNA problems, they're quick to self-destruct, killing off any cells that might one day turn cancerous.

This surplus of TP53 genes is a clear example of how evolution arms large animals with extra layers of defense against tumor growth. TP53 sits at the heart of this system. Studying TP53 in elephants versus humans isn't just a neat comparison—it actually helps explain how evolution shapes cancer resistance, and why some species are simply built to handle the risks better than others.

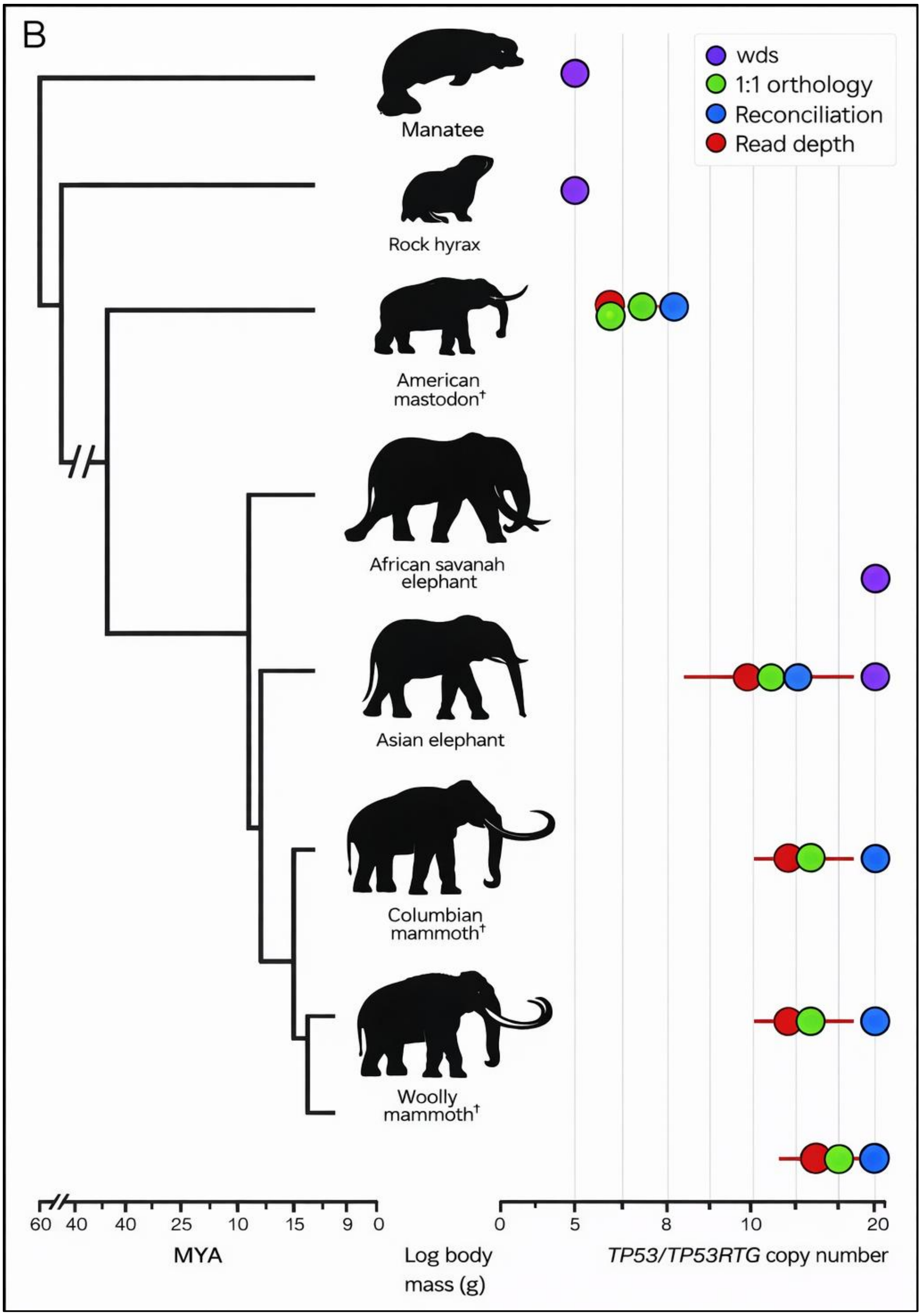

Figure 1.4: Comparative representation of TP53 copy number expansion in elephant species ([4]).

1.5 Rationale of the Present Study

Peto's paradox really changes how we look at cancer across different species. You'd think the bigger the animal and the longer it lives—like elephants or whales—the more likely it is to get cancer. After all, more cells mean more chances for something to go wrong, and a long lifespan gives mutations plenty of time to build up. On paper, these giants should be riddled with cancer. But that's not what we see. Whales and elephants actually dodge cancer far more than expected. They've figured something out.

This gap between what theory says and what actually happens—that's the core of Peto's paradox. It's a sign that big animals evolved powerful ways to keep cancer in check. Over time, natural selection nudged them toward stronger tumor suppression. Their bodies developed systems to keep cell growth under control and protect their DNA, even though they're packed with cells that could go rogue.

Look at elephants. They're a perfect example. Despite their massive size and long lives—more than sixty years, sometimes—they rarely die from cancer. Once scientists started looking at their genes, they found something striking. Elephants have extra copies of the TP53 gene. This gene is key for spotting DNA damage and making sure damaged cells self-destruct before things spiral out of control. These extra TP53 copies come from ancient gene duplications, and they make elephant cells quick to pull the plug if their DNA gets messed up. So, would-be cancer cells don't even get a chance.

The fact that elephants carry all these TP53 genes really backs up the idea that bigger, long-lived animals evolved better tumor suppression to stay ahead of cancer risk. Since TP53 is so important for protecting the genome, comparing how it works in elephants and humans could teach us a lot about how evolution fights cancer.

CHAPTER 2
REVIEW OF LITERATURE

2.1 TP53 as a Central Tumor Suppressor in Human Cancer

Scientists regard the TP53 gene as fundamental to tumor suppression in human cancer biology. Since its discovery, numerous studies have consistently demonstrated the crucial role of TP53 in maintaining genome stability and preventing healthy cells from becoming cancerous. ([1]; [2]). The p53 protein operates as a sequence-specific transcription factor that regulates an extensive network of downstream genes involved in cell cycle arrest, DNA repair, apoptosis, cellular senescence, angiogenesis inhibition, and metabolic regulation ([3]). Through these coordinated mechanisms, TP53 serves as a master regulator of cellular stress responses.

Comprehensive genomic analyses have demonstrated that TP53 is mutated in approximately 50% of all human cancers, making it the most frequently altered gene across diverse malignancies ([4]; [5]). Its mutation frequency is particularly high in solid tumors, including lung carcinoma, colorectal cancer, breast carcinoma, ovarian cancer, and pancreatic cancer. In certain tumor types, TP53 mutation prevalence can exceed 60–70% ([6]). The widespread occurrence of TP53 alterations underscores its indispensable role in tumor suppression and highlights its vulnerability during oncogenic progression.

Importantly, TP53 mutations are not randomly distributed throughout the gene. Instead, they cluster predominantly within the central DNA-binding domain, a region critical for transcriptional regulation ([7]). Specific codons—such as 175, 245, 248, 249, 273, and 282—are recognized as mutation hotspots, where amino acid substitutions frequently impair DNA binding and disrupt transcriptional activation of target genes ([8]). Many of these hotspot mutations result in loss of tumor suppressor function and, in some cases, confer dominant-negative or gain-of-function properties that actively promote tumor progression ([9]).

Functional inactivation of TP53 has profound cellular consequences. Loss of p53-mediated checkpoint control allows cells harboring DNA damage to continue proliferating. Reduced DNA repair efficiency and impaired apoptotic signaling further promote genomic instability, enabling accumulation of additional oncogenic mutations. Over time, this destabilized genomic environment accelerates tumor evolution and contributes to therapeutic resistance ([10]).

Given its structural sensitivity, evolutionary conservation, and dominant presence in cancer mutational landscapes, TP53 remains a central focus of molecular oncology. Ongoing research examines its mutation spectra, structural dynamics, regulatory circuitry, and therapeutic reactivation strategies. In this context, comparative analysis of TP53 mutation hotspots across species—particularly in organisms with unusually low cancer incidence—provides a biologically meaningful framework for understanding evolutionary tumor suppression mechanisms and differential cancer susceptibility.

2.2 Structural Domains of p53 and Functional Consequences of Mutation

Although TP53 mutations occur throughout the coding region, they are not randomly distributed. Instead, mutational events cluster predominantly within the central DNA-binding domain, a structurally and functionally critical region of the p53 protein ([1]). This domain is responsible for sequence-specific recognition of p53 response elements within target gene promoters and is essential for transcriptional activation of genes involved in tumor suppression.

Large-scale mutation databases such as COSMIC, TCGA, and the IARC TP53 Database consistently identify recurrent mutation hotspots at specific codon positions, including R175, G245, R248, R249, R273, and R282 ([2]). These codons are mutated at disproportionately high frequencies across multiple cancer types ([3]). The recurrence of these mutations suggests strong positive selection during tumor evolution, as alterations at these positions significantly impair p53 function.

Structurally, hotspot residues are located at positions that are either directly involved in DNA contact or critical for maintaining structural stability of the DNA-binding domain ([4]). Two principal classes of hotspot mutations have been described. Contact mutations, such as those affecting R248 and R273, directly disrupt interactions between p53 and DNA without necessarily destabilizing the overall protein structure. In contrast, structural mutations, including R175 and G245, alter the folding stability of the DNA-binding domain, leading to conformational instability and loss of functional integrity ([5]).

The functional consequences of these mutations are profound. Impaired DNA binding reduces transcriptional activation of downstream target genes responsible for cell cycle arrest and apoptosis. Additionally, certain mutant forms of p53 exhibit dominant-negative behavior or gain-of-function properties that actively promote oncogenic signaling, invasion, and resistance to therapy ([6]).

The concentration of mutations within structurally sensitive regions underscores the evolutionary and functional importance of these residues. Their high recurrence frequency across diverse malignancies reflects selective advantage in tumor progression. Consequently, these hotspot codons provide a focused framework for comparative residue-level analysis across species.

2.3 TP53 Mutation Databases and Hotspot Codon Identification

Recent advances in high-throughput sequencing technologies have enabled comprehensive and systematic characterization of somatic mutations across diverse cancer types ([1]). Large-scale genomic initiatives have generated extensive datasets cataloging the mutational landscape of cancer-associated genes, with TP53 emerging as the most frequently mutated gene in human malignancies ([2]). Several curated databases have played a pivotal role in consolidating this information, including the Catalogue of Somatic Mutations in Cancer (COSMIC), The Cancer Genome Atlas (TCGA), and the International Agency for Research on Cancer (IARC) TP53 Database ([3]; [4]). These repositories integrate mutation data from thousands of tumor samples across multiple cancer types, providing detailed information on mutation frequency, codon distribution, and functional consequences.

Analysis of these large datasets consistently reveals that TP53 mutations are not randomly distributed along the coding sequence. Instead, specific codons exhibit disproportionately high mutation frequencies and are recognized as recurrent "hotspots" ([5]). Among the most frequently mutated residues are R175, G245, R248, R249, R273, and R282. These codons are located predominantly within the central DNA-binding domain and correspond to structurally or functionally critical residues required for sequence-specific DNA interaction or maintenance of protein conformation ([6]).

The recurrence of mutations at these hotspot positions across independent tumor types suggests strong positive selection during tumorigenesis ([7]). Mutations at contact residues such as R248 and R273 directly impair DNA binding, whereas mutations at structural residues such as R175 and G245 destabilize the protein's folding architecture ([8]). The high prevalence of these mutations indicates that disruption of p53's transcriptional regulatory function provides a selective growth advantage to malignant cells.

Importantly, mutation frequency patterns vary across cancer types, reflecting both environmental exposures and tissue-specific selective pressures. For example, particular TP53 mutations are associated with carcinogen exposure, such as tobacco-derived mutagens in lung cancer or ultraviolet radiation in skin cancer ([9]). These context-dependent mutation signatures further highlight the interplay between environmental factors and structural vulnerability within the TP53 gene.

The well-characterized distribution of TP53 hotspot codons provides a focused and biologically meaningful framework for comparative evolutionary analysis. Because these residues represent functionally critical and structurally sensitive regions of the protein, examining their conservation across species offers valuable insight into evolutionary constraints and tumor suppression mechanisms. This foundation supports the rationale for targeted comparative analysis of hotspot codons between humans and elephants.

2.4 Peto's Paradox and Evolutionary Perspectives on Cancer Resistance

Peto's paradox describes the counterintuitive observation that cancer incidence does not increase proportionally with body size or lifespan across species ([1]). From a theoretical standpoint, organisms with larger body mass possess vastly greater numbers of cells and typically undergo more total cell divisions over their lifetime. Each cell division presents an opportunity for replication errors and mutation accumulation. Accordingly, larger and longer-lived organisms would be expected to exhibit substantially higher cancer rates than smaller, short-lived species. However, comparative epidemiological data do not support this prediction. Large mammals such as elephants and whales do not demonstrate disproportionately elevated cancer incidence, challenging simplistic mutation-based models of cancer risk ([2]).

The existence of this paradox has stimulated extensive research into species-specific tumor suppression mechanisms. If larger organisms do not experience dramatically increased cancer burden, they must possess compensatory biological adaptations that enhance genomic protection. Evolutionary pressures would strongly favor such adaptations, as unchecked cancer mortality would negatively impact reproductive success and survival in large-bodied species ([3]).

Elephants have emerged as one of the most compelling models for investigating the resolution of Peto's paradox. Despite their substantial body mass and lifespan exceeding six decades, elephants exhibit comparatively low cancer mortality rates ([4]). This observation suggests the presence of reinforced tumor suppression pathways. Genomic studies have identified expansion of TP53 gene copies in elephants, including multiple retrogenes derived from ancestral duplication events ([5]). These additional gene copies are believed to enhance the cellular response to DNA damage, increasing the likelihood of apoptosis in cells harboring genomic instability ([6]).

From an evolutionary perspective, large-bodied species likely evolved multiple layers of genomic safeguarding mechanisms to compensate for increased cellular burden. These adaptations may include gene duplication events, improved DNA repair efficiency, enhanced apoptotic sensitivity, strengthened cell cycle checkpoints, and refined stress response pathways ([7]). Such mechanisms collectively reduce the probability that damaged cells will persist and accumulate oncogenic mutations.

Understanding the evolutionary solutions to Peto’s paradox provides valuable insight into natural cancer resistance. Comparative analysis of tumor suppressor genes, particularly TP53, across species offers a biologically meaningful framework for investigating how genomic architecture influences cancer susceptibility. This evolutionary lens strengthens the rationale for examining TP53 hotspot conservation between humans and elephants in the present study.

2.5 TP53 Copy Number Expansion in Elephants

Genomic investigations have demonstrated that elephants possess multiple copies of the TP53 gene, including both the canonical functional gene and several retrogene variants ([1]). Retrogenes are generated through reverse transcription of mature mRNA followed by reintegration into the genome. Although many retrogenes in various organisms are nonfunctional pseudogenes, a subset of elephant TP53 retrogenes appears to retain transcriptional activity and contribute to cellular stress responses ([2]; [3]). This unusual expansion of TP53 copy number distinguishes elephants from most other mammals, including humans, which typically harbor a single canonical TP53 gene.

Experimental and comparative genomic studies have suggested that TP53 copy number expansion in elephants may enhance cellular sensitivity to DNA damage ([4]). Functional assays have shown that elephant fibroblasts display an increased propensity to undergo apoptosis following genotoxic stress when compared to human cells. This heightened apoptotic response likely facilitates rapid elimination of cells harboring genomic instability, thereby reducing the probability of malignant transformation. Such reinforced DNA damage surveillance mechanisms may provide a mechanistic explanation for the comparatively low cancer incidence observed in elephants.

The presence of multiple TP53 copies is therefore hypothesized to increase the robustness and redundancy of tumor suppression pathways. From an evolutionary standpoint, gene duplication events may have been selectively advantageous in large-bodied species by strengthening genomic safeguarding systems and mitigating the increased mutational burden associated with greater cellular mass ([5]). This expansion offers a compelling molecular explanation for Peto’s paradox in elephants.

However, most existing studies have focused primarily on gene copy number, transcriptional activity, and cellular apoptotic responses. Limited attention has been directed toward detailed residue-level comparison of structurally and functionally critical TP53 mutation hotspots between humans and elephants. In particular, the conservation or divergence of canonical human hotspot codons—such as R175, G245, R248, R249, R273, and R282—within elephant canonical and retrogene sequences remains insufficiently explored in a systematic comparative framework.

This knowledge gap underscores the need for precise alignment-based, codon-level comparative analysis to assess whether established TP53 hotspot residues are evolutionarily conserved across species or display substitutions that could influence structural stability, DNA-binding efficiency, or overall tumor suppressor functionality.

2.6 Comparative Genomics and Codon-Level Conservation Analysis

Comparative genomics provides a powerful framework for identifying conserved and divergent genetic elements across species ([1]). By aligning homologous protein sequences, researchers can assess evolutionary constraints acting on specific residues and domains. Conservation across distantly related species often reflects structural or functional indispensability, whereas variability may indicate relaxed selective pressure, adaptive divergence, or lineage-specific functional modification ([2]).

Residue-level conservation analysis is particularly informative for proteins with well-characterized structural and functional domains. Amino acid positions that are highly conserved across evolutionary timescales typically contribute to catalytic activity, structural integrity, molecular interactions, or regulatory control ([3]). Conversely, substitutions at non-critical positions may be tolerated without substantial functional disruption. Therefore, codon-specific comparison allows targeted evaluation of biologically significant residues within a protein.

In the context of TP53, codon-level conservation analysis offers a focused strategy for examining mutation-prone residues associated with human cancers. Well-established hotspot codons such as R175, G245, R248, R249, R273, and R282 are known to play critical roles in maintaining structural stability and DNA-binding capability ([4]; [5]). Mapping these positions onto homologous TP53 sequences from elephants enables direct assessment of whether these functionally critical residues are evolutionarily conserved or exhibit divergence among canonical and retrogene copies.

Such analysis provides insight into evolutionary constraint and structural preservation within the DNA-binding domain. If hotspot residues remain conserved across species and gene copies, this would indicate strong selective pressure to maintain functional integrity. Conversely, divergence at these positions could suggest adaptive modifications or functional diversification within retrogene variants.

Despite major advances in comparative genomics and the availability of high-quality genome assemblies, comprehensive codon-level mapping of canonical human TP53 hotspot residues onto elephant TP53 sequences remains insufficiently explored ([6]).

Previous investigations have largely focused on TP53 copy number expansion and enhanced apoptotic responsiveness in elephants, with limited emphasis on detailed residue-specific alignment of functionally critical hotspot positions. This leaves an important analytical gap at the level of structural and functional codon comparison. Consequently, alignment-based hotspot mapping represents a focused and biologically relevant strategy to evaluate evolutionary conservation within a key tumor suppressor gene and to assess whether critical DNA-binding residues are maintained or diverge across species with distinct cancer susceptibilities.

2.7 Limitations of Existing Research and Identified Knowledge Gap

Although TP53 mutation hotspots in human cancers and TP53 copy number expansion in elephants have been extensively investigated as independent research domains, integration of these two areas remains limited. Human-centric studies primarily focus on mutation frequency distribution, structural destabilization within the DNA-binding domain, functional loss, gain-of-function effects, and therapeutic targeting strategies ([1]; [2]). These investigations have generated comprehensive catalogs of recurrent hotspot codons and clarified their mechanistic contributions to tumorigenesis. In contrast, elephant-focused studies predominantly emphasize TP53 gene copy number expansion, transcriptional activity of retrogenes, and enhanced apoptotic responses following DNA damage ([3]; [4]). While these findings provide compelling explanations for cancer resistance in elephants, they rarely extend to residue-level comparative analysis of structurally critical positions.

Despite the availability of complete protein sequences and advanced alignment methodologies, few investigations have systematically examined whether canonical human TP53 hotspot codons are conserved, modified, or diversified within elephant canonical and retrogene sequences. The absence of structured codon-specific comparative mapping limits our understanding of evolutionary constraints operating at functionally critical residues. Moreover, computational frameworks designed specifically to organize and interpret hotspot-centered residue comparisons across species are not widely implemented in comparative oncology research.

This gap underscores the need for a focused in-silico comparative approach that integrates human cancer mutation data with cross-species sequence analysis. By performing codon-level mapping of biologically significant TP53 residues across human and elephant sequences, it becomes possible to assess evolutionary conservation within mutation-prone structural regions. Such integration not only strengthens comparative genomics methodologies but also enhances interpretability by directly linking well-characterized cancer hotspots to evolutionary tumor suppression strategies.

Therefore, the present study seeks to bridge this conceptual and methodological divide by implementing a systematic alignment-based framework for hotspot mapping. This approach provides evolutionary insight while maintaining analytical transparency, reproducibility, and structural precision.

CHAPTER 3
AIM, OBJECTIVES AND SCOPE

3.1 Aim of the Study

This study introduces EleProtect, a computational framework built to compare codon-level details between well-known TP53 mutation hotspots in humans and the matching sites in both canonical and retrogene TP53 sequences from elephants. The team brings together curated human cancer hotspot data, cross-species protein alignments, detailed residue mapping, and feature engineering. Their goal? To see how much these vital TP53 sites—especially the ones that matter for function—are conserved or have changed, and what those changes mean for the protein's structure. Using alignment-based mapping, substitution scoring, and computational modeling, the researchers get right to the heart of the issue: Are the DNA-binding and structural hotspot residues that are critical in humans also present in elephants? By answering this, the study sheds light on how tumor suppression works differently between species, all through a pipeline built to be reproducible and ready for whatever comes next.

3.2 Objectives of the Study

To achieve the stated aim of conducting a codon-level comparative analysis of TP53 mutation hotspots between humans and elephants, the following objectives were defined:

1. To curate and compile established human TP53 mutation hotspot codons from publicly available cancer mutation databases and peer-reviewed literature.
2. To retrieve and preprocess canonical and retrogene TP53 protein sequences from African and Asian elephants using standardized bioinformatics resources.
3. To perform protein-level multiple sequence alignment between human TP53 and elephant TP53 sequences in order to establish positional correspondence across species.
4. To perform protein-level multiple sequence alignment between human TP53 and elephant TP53 sequences in order to establish positional correspondence across species.
5. To implement alignment-based codon-level mapping of selected human TP53 hotspot residues onto elephant canonical and retrogene sequences.
6. To annotate mapped hotspot positions with functional classifications (e.g., structural vs DNA-contact hotspots).
7. To develop a descriptive computational scoring framework integrating:

    a. Cross-species conservation
    b. Mutation frequency data
    c. Substitution scoring (e.g., BLOSUM62)
    d. Retrogene variability

8. To construct a reproducible computational workflow (EleProtect) capable of automated sequence input, hotspot mapping, and structured output generation.
9. To construct a reproducible computational workflow (EleProtect) capable of automated sequence input, hotspot mapping, and structured output generation.
10. To provide a modular platform extensible for future structural, functional, and cross-species tumor suppression studies.

3.3 Scope of the Study

This study focuses on a select group of well-known TP53 mutation hotspots in humans, aligning them codon by codon with the corresponding sites in both canonical and retrogene TP53 sequences from elephants. The analysis centers on protein sequences—aligning them, mapping specific amino acids, assessing conservation at each site, and utilizing computational scoring.

The scope is narrow. Only the primary hotspot codons within TP53's DNA-binding domain, as identified in leading human cancer mutation databases, are examined. For elephants, the analysis is limited to African and Asian species, using only publicly available genomic databases. All procedures are computational—sequence preparation, global alignments, extraction of hotspot regions, scoring each substitution, and conducting preliminary modeling based on observed patterns.

There is no experimental component—no wet lab validation, no structural modeling, no transcriptomics or gene expression analysis, and no direct correlation of results to clinical outcomes. All functional interpretations rely exclusively on existing biological annotations and substitution scoring matrices.

The computational platform used, EleProtect, is designed to be modular and reproducible. It is adaptable—future research can add new species, target additional hotspots, or incorporate structural and experimental data. Within these parameters, the study provides a focused evolutionary comparison of key TP53 residues between species with markedly different cancer susceptibilities.

CHAPTER 4
METHODOLOGY

4.1 Study Design Overview

We began this study with a clear plan: to use only computational tools to compare TP53 mutation hotspots in humans with the TP53 sequences found in elephants. There was no laboratory work involved—just pure sequence analysis. From the outset, we prioritized clarity and reproducibility, adhering strictly to a sequence-based comparative approach.

Our primary question was straightforward: How well are key TP53 hotspot codons conserved across species? To address this, we divided the workflow into clear, manageable steps. Organizing each phase in this way kept the analysis focused and helped us avoid unnecessary confusion. Here is the structure we followed:

1. Retrieval of canonical TP53 protein sequences from humans and elephants
2. Collection and curation of human TP53 hotspot mutation data
3. Compilation of elephant TP53 retrogene sequences
4. Multiple sequence alignment (MSA) at the protein level
5. Codon-level positional mapping of hotspot residues
6. Structured comparative and descriptive analysis

We designed this study as a focused comparison, not a broad look across many species. To keep everything aligned, we used the human TP53 canonical sequence for codon numbering and structural mapping. Then we brought in elephant TP53 canonical sequences, along with any known retrogene variants, to see how these extra gene copies stack up in terms of conservation.

Hotspot codons weren't chosen on a whim. We leaned on mutation frequency data from well-established cancer mutation databases and highly-cited peer-reviewed studies. Only hotspot residues with strong research and clear structural roles made our list. That kept our analysis sharp and meaningful.

For the alignments, we skipped the nucleotide level and went straight to proteins. This way, we preserved the structural context and avoided the noise from synonymous changes. So, when we compared residues, we really looked at the functional amino acid positions in the DNA-binding domain.

We mapped human TP53 hotspot codons onto the elephant sequences and checked for conservation at those key sites. Everything ran computationally, using publicly available genomic data. What we're presenting is a streamlined, in-silico approach for comparing TP53 hotspot residues across species.

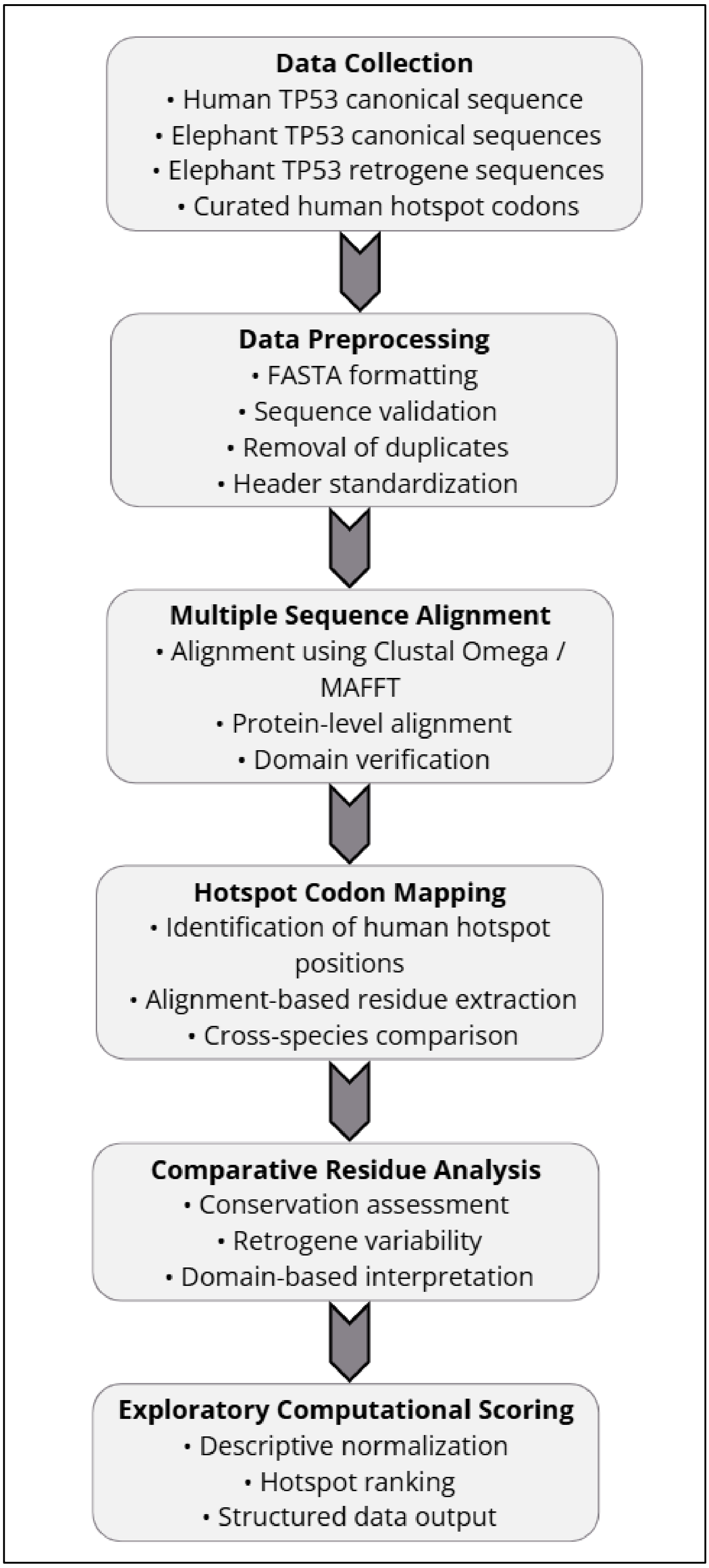

Figure 2.1: Workflow of the in-silico comparative TP53 analysis performed in this study.

4.2 Data Collection

4.2.1 Human TP53 Sequence Retrieval

I obtained the human TP53 protein sequence directly from UniProt and NCBI, selecting the standard isoform commonly used for identifying mutation hotspots. This ensures that my numbering matches the major cancer mutation databases. I downloaded the sequence in FASTA format, which facilitates its use in alignment tools and other bioinformatics analyses. After confirming that the sequence was complete and correctly annotated, I used it as the reference for mapping hotspot codons onto the elephant TP53 sequences.

4.2.2 Elephant TP53 Canonical Sequences

I directly retrieved the canonical TP53 protein sequences for African and Asian elephants from public genome databases—NCBI and Ensembl. I didn't just select random entries. I only included complete, well-annotated sequences without any coding gaps. To ensure everything aligned with the human reference, I concentrated on thoroughly documented canonical proteins.

All sequences were saved in FASTA format, as it works best with the alignment and analysis tools I use. I organized everything into project folders, which is essential once you start preprocessing, running alignments, and mapping codons. Before comparing sequences, I double-checked the lengths and annotation details. It's better to catch any inconsistencies early on rather than moving forward with errors.

4.2.3 Elephant TP53 Retrogene Sequences

We obtained elephant TP53 retrogene sequences from public genomic databases and verified them against published studies. For each sequence, we ensured the coding regions were complete and correctly annotated. When nucleotide sequences were available, we translated them into protein sequences to maintain consistency for protein-level alignment.

We checked the open reading frames for errors and excluded any sequences that were truncated or appeared incorrect. Each retrogene was assigned a standard ID, making it easy to track them during multiple sequence alignment and codon-level comparisons.

4.2.4 Human TP53 Hotspot Codon Data

Curated TP53 mutation hotspot codon information was compiled from established cancer mutation databases, including COSMIC and TCGA summaries, along with validated peer-reviewed literature sources. These repositories were used to identify recurrent and functionally significant mutation sites within the TP53 coding sequence.

Hotspot codon selection was based on defined criteria:

a. High mutation frequency across multiple cancer types
b. Localization within functionally significant domains
c. Recurrence reported in curated mutation databases

Based on these criteria, key hotspot residues selected for analysis included R175, G245, R248, R249, R273, and R282. The curated hotspot information was systematically organized into structured tabular format to facilitate integration with sequence alignment and codon-level mapping analysis.

4.3 Data Preprocessing

All downloaded sequences underwent preprocessing to ensure consistency, accuracy, and suitability for comparative alignment. This quality-control stage was essential to minimize analytical errors arising from incomplete or improperly annotated entries.

Preprocessing steps included:

a. Removal of incomplete sequences
b. Verification of correct reading frame
c. Removal of duplicate entries
d. Standardization of FASTA headers

We used protein sequences for multiple sequence alignment so we could accurately compare residues across different species. Working at the protein level maintained a clear connection between codons and amino acids, and we avoided the complications of intron–exon boundaries present in genomic DNA. Genomic sequences often introduce additional noise—non-coding regions, alternative splicing, annotation errors—that make it more difficult to identify positions. By focusing on translated proteins, we concentrated on the actual functional amino acids, particularly within the critical domains of p53.

Aligning at the protein level also avoids confusion from synonymous substitutions—nucleotide changes that do not alter the amino acid but can complicate nucleotide alignments. Since our aim was to investigate conservation and divergence at mutation-prone hotspot residues, comparing amino acid sequences provided a more direct and meaningful perspective.

Throughout the study, we organized all sequence files in a clear, hierarchical directory structure, with raw data, processed sequences, and analysis results each stored separately. This organization made everything easier to manage, ensured our work could be reproduced by others, and kept our workflow efficient from start to finish.

4.4 Multiple Sequence Alignment

Working at the protein level cuts through the mess you get with genomic DNA—no introns, no headaches from alternative splicing, and no fighting with inconsistent annotations. Genomic sequences are packed with non-coding regions and random quirks that make alignment a real challenge. By translating everything to protein, I zeroed in on amino acids that actually matter, so I could compare the real, functional parts of the p53 protein.

Focusing on proteins also wipes out the background noise from synonymous substitutions. Those changes in DNA don't touch the amino acid sequence, but they turn DNA alignments into a confusing puzzle and muddy the biological picture. Since I needed to see how mutation-prone hotspots in the DNA-binding domain either stick around or change across species, protein alignment gave me exactly what I needed—a clear way to compare the regions that count. It links hotspot codons straight to the amino acids that drive function, not just some random spot on the genome.

Using protein sequences kept everything lined up with the standard human TP53, which I used as my reference for mapping hotspots. Sticking to the same numbering system meant I could confidently pinpoint matching residues in both the canonical elephant sequence and the retrogenes.

To keep things transparent and reproducible, I set up all my sequence files in a tidy directory. The raw data folders held the original FASTA files straight from the databases. Processed data folders stored the cleaned-up sequences, ready for alignment. Results folders kept the alignment outputs, all the figures, phylogenetic trees, and hotspot summaries in one place.

This setup made the workflow easy to follow, cut out any duplicate work, and left me with a clear trail for every step of the analysis.

4.5 Domain Annotation

We mapped out the functional domains of human TP53 using well-established structural and biochemical research. p53 isn't a straightforward protein. It's made up of several domains, each critical for its role as a tumor suppressor: the N-terminal transactivation domain, a proline-rich segment, the central DNA-binding domain, the tetramerization domain, and the C-terminal regulatory region. For this study, we zeroed in on the transactivation, DNA-binding, and tetramerization domains, since these connect directly to mutation hotspots and p53's main functions.

To set domain boundaries, we used structural data from experiments and top database annotations. We then mapped these domains onto the human TP53 reference sequence and extended this mapping to both canonical and retrogene sequences in elephants using a multiple sequence alignment. This let us pinpoint exactly where hotspot residues fall within the structural framework.

Careful domain annotation turned out to be crucial. Most cancer-associated mutations cluster in the central DNA-binding domain—the part that determines how p53 recognizes its target genes. By clearly marking domain limits in our alignment, we could see if each hotspot residue landed inside a structurally sensitive zone or outside the protein's main functional regions.

Anchoring each codon in its structural context gave us a stronger way to compare across species. We could tie codon conservation patterns directly to the protein's most important regions. In short, domain annotation sharpened our interpretation: it let us see residue conservation as true preservation of these key structural areas, not just a matter of matching sequences.

4.6 Hotspot Codon Mapping Strategy

The hotspot codon mapping strategy constituted the central analytical component of this study. Following completion and verification of the multiple sequence alignment (MSA), human TP53 hotspot codon positions were systematically mapped onto the aligned sequence framework to enable residue-level comparison across species.

The canonical human TP53 sequence was used as the positional reference for codon numbering. For each selected hotspot codon (R175, G245, R248, R249, R273, and R282), the corresponding amino acid position was first identified within the human reference sequence. Care was taken to ensure that alignment gaps did not disrupt positional accuracy. Manual inspection of alignment columns was performed where necessary to confirm correct positional correspondence.

Once the human hotspot position was located within the alignment, the equivalent aligned positions in African elephant canonical TP53, Asian elephant canonical TP53, and available TP53 retrogene sequences were extracted. Each residue present at that aligned column was recorded for all sequences included in the study.

Residue identity was documented in structured tabular format. For each hotspot codon, the following information was recorded:

1. Human reference residue
2. Elephant canonical residue(s)
3. Elephant retrogene residue(s)
4. Presence or absence of substitution

This codon-level mapping approach enabled precise cross-species comparison at biologically significant residues associated with cancer mutagenesis in humans. By focusing exclusively on predefined hotspot positions, the analysis remained targeted and biologically relevant.

Mapping results were stored in structured datasets to enable reproducible analysis, graphical representation, and potential downstream computational integration. This systematic mapping framework ensured analytical transparency and minimized interpretive ambiguity.

4.7 Comparative Residue Analysis

Following codon-level mapping, comparative residue analysis was conducted to evaluate conservation and divergence patterns across species and gene copies. The analysis focused on three primary parameters: residue identity across species, conservation between human and elephant canonical sequences, and variation among elephant TP53 retrogene copies.

For each mapped hotspot codon, residue identity was first assessed between the human reference sequence and elephant canonical sequences. Complete identity at a given position was interpreted as strong evolutionary conservation, suggesting functional constraint at that residue. Conservation at structurally critical residues within the DNA-binding domain was considered particularly significant due to their role in maintaining p53 structural stability and DNA-binding capability.

The second parameter involved evaluating variation among elephant retrogene copies. Because retrogenes arise through duplication events, divergence among copies may indicate relaxed selective pressure or functional diversification. Positions where retrogene sequences retained identical residues to the canonical sequence were interpreted as conserved, whereas substitutions were documented as potential divergence events.

Importantly, no structural modeling, molecular dynamics simulations, or protein stability prediction algorithms were applied in this study. The analysis remained descriptive and comparative in nature, focusing solely on residue-level conservation patterns within a sequence-based framework.

This comparative residue analysis provided evolutionary context to human cancer hotspot positions while maintaining methodological simplicity and transparency.

4.8 Exploratory Computational Scoring Approach

To organize hotspot comparison results in a structured manner, an exploratory computational scoring approach was implemented. This scoring system was strictly descriptive and was not intended for predictive, diagnostic, or clinical interpretation.

Three primary features were incorporated into the scoring framework:

1. Reported mutation frequency of each hotspot codon in human cancers
2. Conservation score across species (human and elephant canonical sequences)
3. Degree of variability among elephant retrogene copies

Mutation frequency data were obtained from curated cancer mutation databases. Conservation scores were assigned based on residue identity patterns observed in the alignment. Retrogene variability was evaluated by assessing whether retrogene copies exhibited substitutions at the mapped hotspot positions.

To ensure comparability across features with different scales, normalization was performed using standard scaling techniques. Each feature was transformed to a comparable range, and an aggregate descriptive score was calculated by averaging normalized feature values.

The resulting score provided a relative ranking of hotspot codons based on mutation prevalence and evolutionary conservation. It did not imply functional impact, tumor suppression strength, or evolutionary advantage. Instead, it served as a structured method to organize and summarize comparative hotspot data.

This exploratory scoring framework enhanced analytical clarity while remaining within defined non-predictive boundaries.

4.9 Statistical and Descriptive Analysis

The study employed descriptive comparative analysis rather than inferential statistical testing. Because the investigation focused on a predefined set of biologically significant hotspot residues rather than large-scale genomic variation, formal hypothesis testing was not conducted.

Conservation patterns were summarized using structured identity comparison tables that documented residue presence across species and gene copies. These tables provided clear visualization of conserved versus substituted positions.

Alignment-based visualization tools were also utilized to support descriptive interpretation. Sequence logos were generated to illustrate residue conservation patterns across aligned sequences. Phylogenetic trees constructed from aligned TP53 sequences provided contextual insight into evolutionary relationships among canonical and retrogene copies.

No statistical modeling of mutation frequency distributions, survival analysis, or clinical outcome prediction was performed. The analytical framework remained focused on structural and evolutionary comparison rather than quantitative inference.

This descriptive strategy ensured that conclusions remained proportionate to the scope of the study and avoided overinterpretation of limited comparative data.

4.10 Reproducibility and Data Organization

Reproducibility was a fundamental consideration throughout the study. All sequence files, processed datasets, alignment outputs, and comparative tables were systematically organized within a clearly defined hierarchical directory structure.

Raw input data were preserved in separate folders to maintain integrity of original downloaded sequences. Processed datasets were stored independently to allow traceability of preprocessing steps. Alignment files, graphical outputs, phylogenetic trees, and hotspot mapping summaries were maintained within dedicated results directories.

Each computational step, including sequence retrieval, preprocessing, alignment, and hotspot mapping, was documented to ensure analytical transparency. File naming conventions were standardized to avoid ambiguity during cross-referencing and result interpretation.

The structured organizational framework allows future researchers to replicate the workflow using updated sequence datasets, incorporate additional species, or extend the codon-level mapping strategy to other tumor suppressor genes.

By maintaining strict separation between raw data and processed outputs, the study adheres to reproducibility standards in computational genomics and supports methodological clarity.

CHAPTER 5
IMPLEMENTATION

5.1 Computational Architecture

The EleProtect system was implemented as a structured and modular computational framework designed to perform comparative TP53 hotspot analysis across species. The architecture integrates sequence preprocessing, alignment-based codon mapping, feature engineering, exploratory machine learning, and interactive web-based deployment into a unified analytical pipeline.

The computational design follows a layered architecture model consisting of four primary components:

1. **Input and Preprocessing Layer**: Responsible for accepting multi-format sequence inputs (FASTA, TXT, DOCX, XLSX, and database accession IDs) and performing sequence cleaning, validation, and automatic nucleotide-to-protein translation.
2. **Alignment and Hotspot Mapping Layer**: Implements global protein alignment using the human canonical TP53 sequence as the positional reference. This layer performs codon-specific extraction of biologically significant hotspot residues.
3. **Feature Engineering and Analytical Layer**: Transforms mapped hotspot results into structured numerical feature vectors including conservation score, mutation frequency, sequence identity, and retrogene variability metrics.
4. **Modeling and Deployment Layer**: Integrates an exploratory machine learning model for structured scoring and deploys the analytical pipeline through a Streamlit-based graphical interface.

The system was initially prototyped in notebook environments (EleProtect_ML_Analysis.ipynb and TP53_AI_Pipeline.ipynb) to evaluate similarity clustering, substitution sensitivity, and feature behavior. Once validated, the logic was modularized into standalone Python scripts (utils.py, model.py, train_model.py) to enable reproducible execution and web-based deployment.

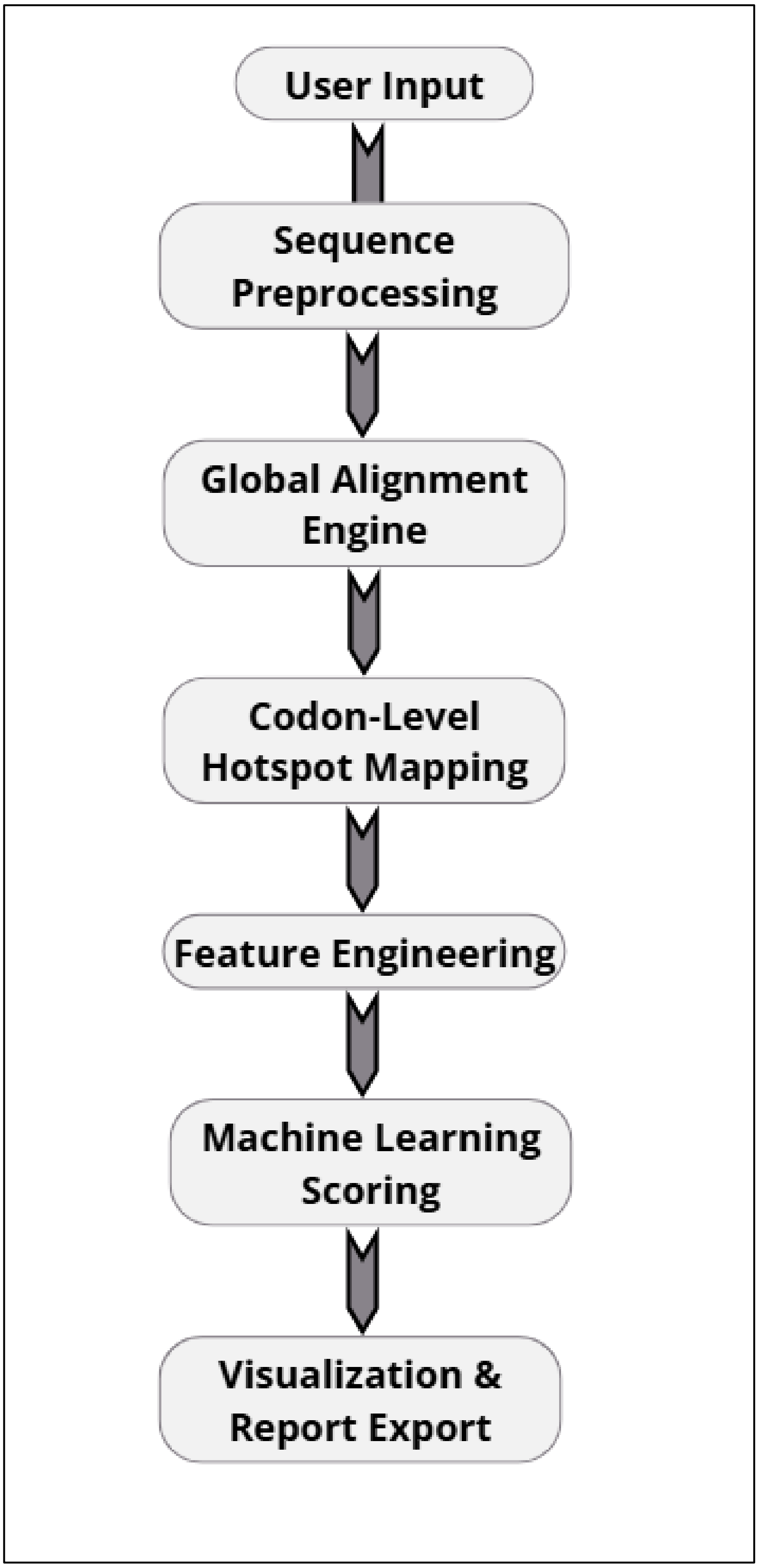

Figure 3.1: Overall Computational Architecture of EleProtect

5.2 Project Directory Structure

The EleProtect framework was organized using a modular and hierarchical directory structure to ensure computational clarity, reproducibility, and separation of concerns. The project architecture distinguishes between core analytical scripts, machine learning components, data resources, serialized models, and exploratory development notebooks.

This structural organization enables independent modification of preprocessing logic, alignment functions, modeling routines, or deployment scripts without disrupting the overall system.

The directory hierarchy is presented below:

```
EleProtect/
 |
 ├── app.py
 ├── utils.py
 ├── model.py
 ├── train_model.py
 ├── generate_training_dataset.py
 |
 ├── models/
 |    ├── eleprotect_model.joblib
 |    └── feature_columns.joblib
 |
 ├── data/
 |    └── tp53_expanded_training_dataset.csv
 |
 └── notebooks/
     ├── EleProtect_ML_Analysis.ipynb
     └── TP53_AI_Pipeline.ipynb
```

Figure 3.2: Modular Directory Architecture of the EleProtect Framework

Structural Components and Functional Roles

1. Core Analytical Scripts

a. **utils.py**

Contains foundational bioinformatics operations including:

i. Sequence cleaning and validation
ii. Nucleotide-to-protein translation
iii. Global alignment execution
iv. Codon-level hotspot extraction
v. Feature engineering

b. **model.py**

Responsible for:

i. Loading serialized machine learning models
ii. Ensuring consistent feature ordering
iii. Performing predictive scoring

c. **train_model.py**

Implements:

i. Model training
ii. Cross-validation
iii. Model serialization

d. **generate_training_dataset.py**

Generates synthetic feature datasets to enhance model robustness in scenarios with limited labeled biological data.

2. Model Storage Directory (models/)

The models/ directory stores serialized machine learning objects using joblib, including:

a. Trained model object
b. Feature column structure

This ensures consistent inference during GUI-based execution and prevents feature mismatch errors.

3. Data Directory (data/)

The data/ directory contains structured training datasets used for supervised learning experiments and validation procedures.

4. Development Notebooks (notebooks/)

The notebooks directory preserves exploratory analysis conducted during development, including:

a. Similarity clustering experiments
b. Feature behavior evaluation
c. Substitution sensitivity testing
d. Model performance experimentation

These notebooks document the analytical reasoning process preceding deployment.

Design Rationale

The directory structure was intentionally designed to:

a. Separate experimentation from production code
b. Support reproducibility and version control
c. Enable model retraining without modifying deployment scripts
d. Maintain clean dependency flow across modules

This modular organization enhances long-term extensibility, allowing integration of additional genes, species, or analytical layers in future work.

5.3 Sequence Preprocessing Implementation

Sequence preprocessing represents the foundational stage of the EleProtect analytical pipeline. Given the diversity of possible input formats—including FASTA files, plain-text sequences, DOCX documents, Excel spreadsheets, and accession-based database retrieval—standardization of input data was essential prior to alignment and hotspot mapping.

The preprocessing module was implemented in utils.py and performs the following operations:

1. Removal of whitespace, line breaks, and non-alphabetic characters
2. Conversion of all characters to uppercase format
3. Detection of nucleotide sequences
4. Automatic translation of nucleotide sequences into protein format

This ensures that all downstream alignment operations are conducted using standardized protein sequences, thereby preventing alignment artifacts caused by formatting inconsistencies.

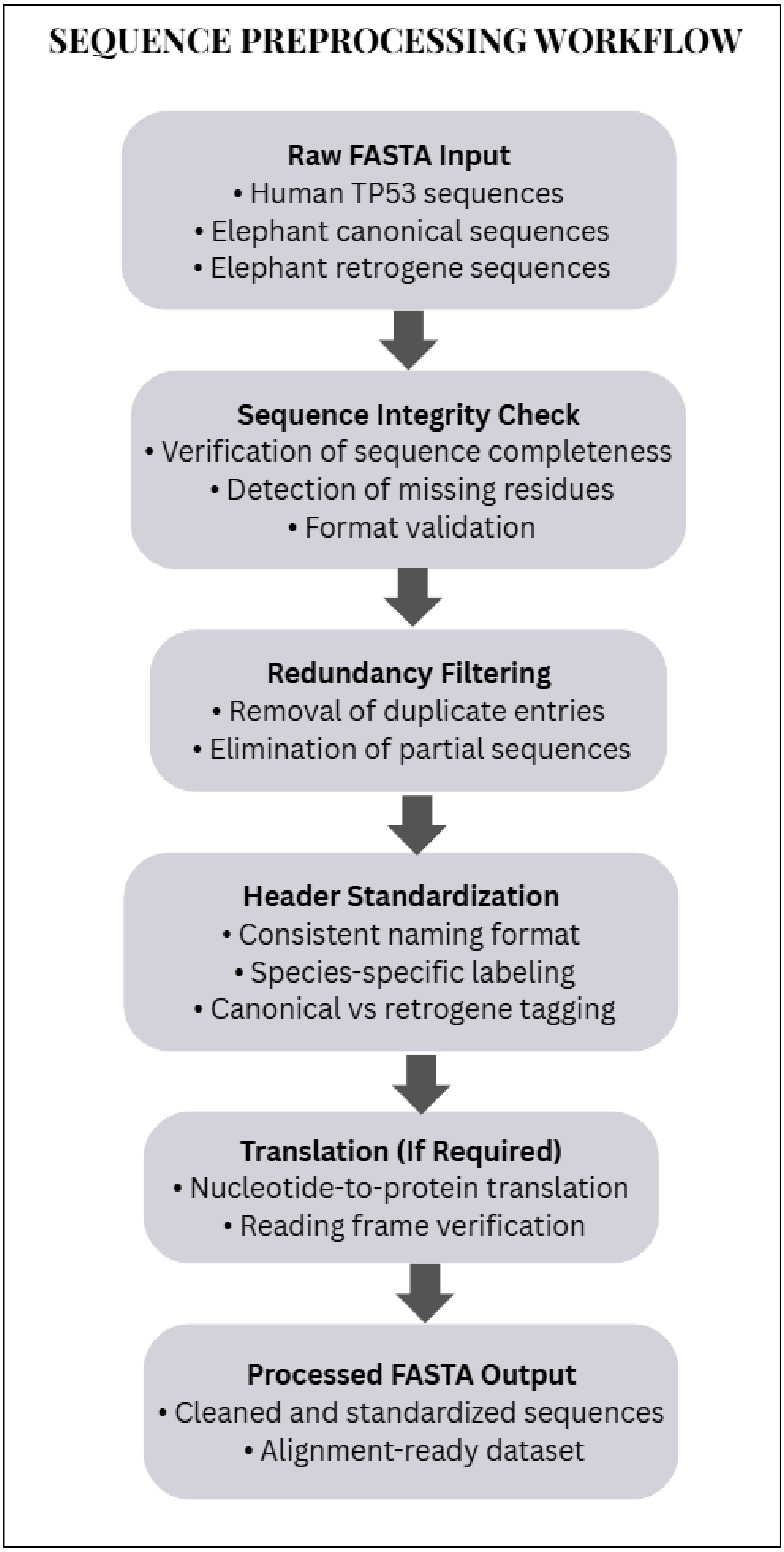

Figure 3.3: Sequence preprocessing workflow illustrating quality control, redundancy removal, header standardisation, and generation of alignment-ready TP53 datasets.

5.3.1. Sequence Cleaning & Translation

Sequence cleaning eliminates extraneous characters introduced during file upload, copying from databases, or formatting artefacts. Only alphabetic characters corresponding to valid amino acids or nucleotides are retained.

To support flexibility in user input, the system includes automatic detection of nucleotide sequences. If the majority of characters belong to the nucleotide alphabet (A, C, G, T, N), translation into protein sequence is performed using the standard genetic code.

Representative Code Snippet (from utils.py)

```python
# =========================================================
# SEQUENCE CLEANING
# =========================================================

def clean_sequence(seq: str) -> str:
    s = str(seq).replace("\r", "").replace("\n", "").strip()
    s = re.sub(r"[^A-Za-z]", "", s)
    return s.upper()


def is_nucleotide(seq: str) -> bool:
    if len(seq) == 0:
        return False
    valid = set("ACGTN")
    return sum(1 for c in seq if c in valid) / len(seq) >= 0.75


def translate_if_needed(seq: str) -> str:
    seq = clean_sequence(seq)
    if is_nucleotide(seq):
        try:
            return str(Seq(seq).translate(to_stop=True))
        except Exception:
            return str(Seq(seq).translate())
    return seq

```

Figure 3.3.1 : Source Code for Sequence Cleaning and Translation Workflow

This function ensures that input sequences are reduced to biologically meaningful characters before further processing.

This logic ensures that:

- Protein sequences remain unchanged
- DNA sequences are converted into biologically interpretable protein format
- Stop codons are handled appropriately

5.3.3 Multi-Format File Parsing

The preprocessing module also supports parsing of multiple file formats through conditional logic that detects file extensions and content structure. FASTA headers are parsed using a header-based split mechanism, while DOCX and Excel files are converted into sequence strings prior to cleaning.

This flexible parsing strategy enables EleProtect to operate in real-world research scenarios where sequence data may originate from heterogeneous sources.

Robustness and Validation

The preprocessing layer includes safeguards against:

- Empty sequence inputs
- Malformed files
- Improper FASTA formatting
- Non-biological character insertion

By enforcing strict cleaning and validation prior to alignment, the system ensures computational stability and reproducibility.

5.4 Multiple Sequence Alignment Execution

Accurate codon-level comparison between human and elephant TP53 sequences requires precise residue alignment. To achieve this, EleProtect implements global protein sequence alignment using the PairwiseAligner class from the Biopython library.

Global alignment was selected because hotspot mapping requires preservation of positional correspondence across the full-length TP53 protein. Since hotspot codons are defined relative to human reference positions, a global alignment strategy ensures consistent indexing of residues across species.

The human canonical TP53 protein sequence serves as the positional reference anchor. All query sequences—including elephant canonical TP53 and retrogene variants—are aligned against this reference prior to hotspot extraction.

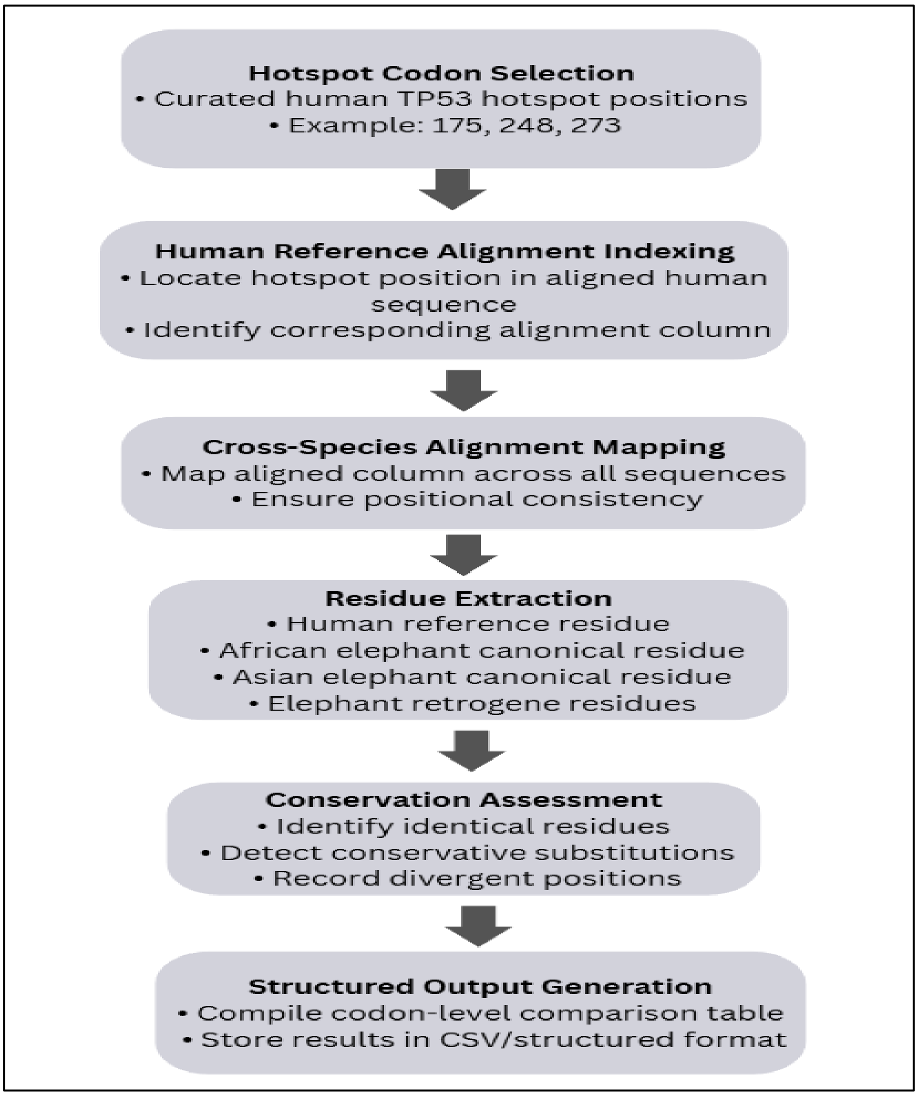

Figure 3.4: Logical workflow for alignment-based mapping of TP53 hotspot codons across human and elephant sequences.

5.4.1. Alignment Parameter Configuration

Alignment scoring parameters were configured to balance match accuracy and gap penalization:

- Match score: +2
- Mismatch penalty: −1
- Gap opening penalty: −10
- Gap extension penalty: −0.5
- Alignment mode: Global

These parameters discourage excessive gap insertion while allowing biologically meaningful substitutions.

Representative Code Snippet (from utils.py)

```python
# ========================================================
# ALIGNMENT & HOTSPOT MAPPING
# ========================================================

def align_and_map(human_seq: str, query_seq: str, hotspots=None) -> pd.DataFrame:

    if hotspots is None:
        hotspots = HOTSPOTS

    if not human_seq or not query_seq:
        return pd.DataFrame()

    aligner = PairwiseAligner()
    aligner.mode = "global"
    aligner.match_score = 2
    aligner.mismatch_score = -1
    aligner.open_gap_score = -10
    aligner.extend_gap_score = -0.5

    alignment = next(iter(aligner.align(human_seq, query_seq)))

    aligned_human = alignment[0]
    aligned_query = alignment[1]

    results = []
    human_pos = 0
```

Figure 3.4.1: Global Alignment Configuration and Hotspot Extraction Logic

This configuration ensures stable full-length alignment while avoiding overflow errors by explicitly selecting only the first optimal alignment.

5.4.2 Alignment Output Handling

The resulting alignment produces two aligned strings:

- Aligned human TP53 sequence
- Aligned query TP53 sequence

These aligned sequences may contain gap characters ("-"), which must be handled carefully during positional mapping. The human sequence is used to track true codon positions by incrementing the positional index only when non-gap residues are encountered.

This step is critical for maintaining accurate codon indexing relative to the canonical human reference sequence.

The alignment strategy implemented in EleProtect is illustrated in Figure 5.4.

5.5 Codon-Level Hotspot Mapping Logic

Following global alignment, the critical task of EleProtect is to extract and compare biologically significant TP53 hotspot residues at predefined codon positions. Unlike general sequence similarity metrics, this step performs targeted codon-level mapping relative to the canonical human TP53 reference.

Hotspot codons analyzed in this study include positions 175, 245, 248, 249, 273, and 282, which are well-documented mutation hotspots within the DNA-binding domain of TP53. These residues were selected due to their high mutation frequency and functional relevance in tumor suppression.

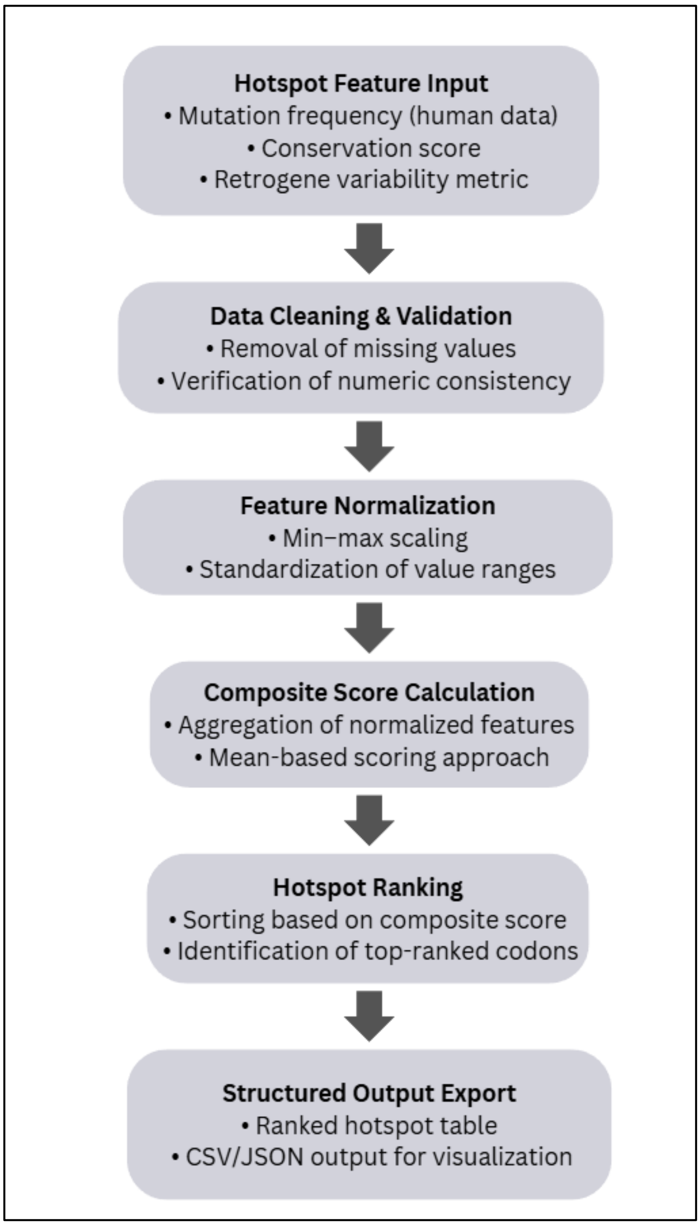

Figure 3.5: Exploratory computational scoring pipeline applied to TP53 hotspot codons based on mutation frequency, conservation, and retrogene variability.

5.5.1. Human Reference Anchoring

Because hotspot positions are defined according to the human TP53 reference sequence, the alignment output must preserve positional accuracy. However, the presence of gaps introduced during alignment complicates direct positional indexing.

To resolve this, EleProtect implements a positional tracking mechanism:

- A counter is incremented only when a non-gap character is encountered in the aligned human sequence.
- When the counter matches a predefined hotspot codon, the corresponding aligned residues from both sequences are extracted.
- Conservation status is evaluated by comparing the two residues.

This ensures that hotspot positions remain biologically meaningful and are not distorted by alignment gap artifacts.

5.5.2 Codon Extraction Algorithm

The following representative code demonstrates the hotspot mapping logic implemented in utils.py.

```
    for i in range(len(aligned_human)):

        if aligned_human[i] != "-":
            human_pos += 1

        if human_pos in hotspots:

            score = blosum_score(aligned_human[i], aligned_query[i])

            results.append({
                "Codon": human_pos,
                "Human": aligned_human[i],
                "Query": aligned_query[i],
                "Conserved": aligned_human[i] == aligned_query[i],
                "BLOSUM62_Score": score
            })

    return pd.DataFrame(results)
```

Figure 3.5.1 : Codon-Level Hotspot Mapping with BLOSUM62 Substitution Scoring.

5.5.3 Conservation Determination

For each hotspot codon:

- If the aligned residues are identical, the position is marked as conserved.
- If residues differ, the position is marked as non-conserved, indicating potential structural or functional divergence.

This binary conservation flag serves as the foundational metric for subsequent feature engineering and scoring.

5.5.4 Biological Annotation Layer

To enhance interpretability, mapped codons are optionally annotated with known biological significance categories (e.g., structural hotspot, DNA-contact mutation). This annotation layer links computational results to established functional knowledge of TP53.

This codon-level mapping approach distinguishes EleProtect from general sequence similarity tools by focusing specifically on functionally critical residues rather than overall alignment identity.

The codon mapping mechanism is illustrated in Figure 5.5.

5.6 Comparative Data Organization

Following hotspot extraction, EleProtect organizes codon-level results into structured tabular formats to enable comparative evaluation across sequences. This step transforms alignment outputs into interpretable comparative genomics data.

5.6.1. Structured Hotspot Mapping Table

For each analyzed sequence, a standardized dataframe is generated with the following fields:

Column	**Description**
Codon	Human TP53 hotspot position
Human	Reference residue
Query	Corresponding residue in analyzed sequence
Conserved	Boolean conservation flag

This tabular structure allows direct residue-level comparison across species or TP53 variants.

A. utils.py

```
# ========================================================
# TRAIN MODEL
# ========================================================

model.fit(X, y)

# ========================================================
# CROSS-VALIDATION
# ========================================================

cv_scores = cross_val_score(model, X, y, cv=5)

print("Cross-validation R2 scores:", cv_scores)
print("Mean CV R2:", np.mean(cv_scores))

# ========================================================
# SAVE MODEL + FEATURE ORDER
# ========================================================

os.makedirs("models", exist_ok=True)

joblib.dump(model, "models/eleprotect_model.joblib")
joblib.dump(FEATURE_COLUMNS, "models/feature_columns.joblib")

print("Model and feature column order saved successfully.")
```

Fig No. 3.6.1 : Model Training, Cross-Validation, and Serialization Workflow

5.6.2 Conservation Percentage Calculation

To summarize residue-level conservation into a comparative metric, a conservation percentage score is calculated as:

$$\text{Conservation}\% = \frac{\text{Number of Conserved Hotspots}}{\text{Total Hotspots}} \times 100$$

This produces a normalized value between 0 and 1, later converted to percentage for reporting.

5.6.3 Batch Processing of Multiple Sequences

EleProtect supports multi-sequence input (FASTA files or database fetches). During batch analysis:

- Each sequence is processed independently.
- Hotspot mapping results are stored.
- Summary statistics are compiled into a unified dataframe.

Representative Code Snippet (from app.py)

```
features = build_model_features(df_map)
score = predict_score(pd.DataFrame([features]))

results.append({
    "Sequence": name,
    "Conservation_%": round(features["conservation_score"]*100,2),
    "ML_Score": round(float(score),4),
    "Mapping": df_map,
    "Length": len(seq_clean)
})
```

Fig No. 3.6.2: Feature Construction and Machine Learning–Based Score Prediction Pipeline

5.6.4 Summary Table Construction

After processing all sequences, a comparative summary table is generated containing:

- Sequence identifier
- Conservation percentage
- Machine learning score

Code Snippet:

```python
# ====================================================
# RESULTS TABS
# ====================================================

st.markdown("---")

tabs = st.tabs(["📊 Summary", "🧬 Hotspot Mapping", "📈 Domain Architecture"])

# SUMMARY TAB
with tabs[0]:
    summary_df = pd.DataFrame([
        {
            "Sequence": r["Sequence"],
            "Conservation_%": r["Conservation_%"],
            "ML_Score": r["ML_Score"]
        }
        for r in results
    ])
    st.dataframe(summary_df, use_container_width=True)

    avg_cons = summary_df["Conservation_%"].mean()
```

Fig No 3.6.3: Streamlit-Based Interactive Results Interface Architecture

5.6.5 Clustering and Similarity Interpretation

In the earlier analytical phase (TP53 similarity clustering notebook), comparative similarity matrices were generated to observe evolutionary grouping patterns among TP53 sequences.

Hierarchical or similarity-based grouping demonstrated:

- Close clustering of elephant canonical TP53 to human reference
- Greater divergence in retrogene variants
- Hotspot-level conservation maintained in key structural residues

This reinforces the biological rationale behind hotspot-focused comparison rather than full-sequence identity alone.

The data organization workflow is illustrated in Figure 5.6.

5.7 Exploratory Computational Scoring Implementation

The scoring module of EleProtect transforms codon-level hotspot mapping results into structured numerical representations suitable for analytical evaluation and exploratory modeling. Rather than relying solely on binary conservation flags, the system derives quantitative features that summarize mutation patterns at biologically significant residues.

This feature-based representation enables both descriptive scoring and integration with machine learning models.

5.7.1. Scoring Implementation

Following hotspot extraction, the mapped dataframe is converted into a structured feature vector. The following metrics are computed:

1. **Conservation Score**
   Fraction of conserved hotspot residues relative to total hotspots analyzed.
2. **Sequence Identity (Hotspot-Level)**
   Equivalent to conservation score but retained as a separate interpretative metric.
3. **Retrogene Variability**
   Defined as: $1 - \text{Conservation Score}$
4. **Mutation Frequency (Curated/Simulated)**
   Represents mutation burden associated with hotspot positions.

These features collectively capture structural conservation and mutation susceptibility patterns.

5.7.2 Feature Vector Construction

The feature construction logic is implemented in utils.py.

```python
# ============================================================
# ADVANCED FEATURE ENGINEERING
# ============================================================

def build_model_features(df_hotspots: pd.DataFrame) -> dict:

    if df_hotspots.empty:
        return {
            "conservation_score": 0,
            "mutation_frequency": 0,
            "retrogene_variability": 0,
            "sequence_identity": 0,
            "mean_blosum": 0,
            "std_blosum": 0,
            "damaging_fraction": 0,
            "hotspot_count": 0,
            "weighted_mutation_burden": 0
        }

    identity = df_hotspots["Conserved"].mean()
    mean_blosum = df_hotspots["BLOSUM62_Score"].mean()
    std_blosum = df_hotspots["BLOSUM62_Score"].std()

    damaging_fraction = (
        (df_hotspots["BLOSUM62_Score"] < 0).sum()
        / len(df_hotspots)
    )

    freq_vals = df_hotspots["Codon"].map(
```

Fig No. 3.7.1: Biologically Informed Feature Engineering Pipeline for Hotspot-Based Predictive Modeling

This ensures consistent feature structure across all analyzed sequences.

5.7.3 Descriptive Composite Scoring

In addition to machine learning-based inference, EleProtect includes a fallback composite scoring mechanism using weighted contributions from engineered features. This ensures interpretability even when model loading fails.

Example weighted formulation:

$$\text{Score} = (0.6 \times \text{Conservation}) - (0.3 \times \text{MutationFrequency}) + (0.2 \times \text{Variability})$$

This weighted scoring approach emphasizes structural conservation while penalizing mutation burden.

5.7.4 Model-Based Predictive Scoring

For enhanced analytical depth, a trained Gradient Boosting Regressor model is applied to the feature vector. The trained model is serialized and loaded during runtime for consistent inference.

Prediction Snippet (from model.py)

```
    freq_vals = df_hotspots["Codon"].map(
        TP53_MUTATION_FREQUENCY
    ).fillna(0)

    mutation_frequency = freq_vals.mean()

    weighted_mutation_burden = (
        freq_vals * abs(df_hotspots["BLOSUM62_Score"])
    ).mean()

    conservation_score = mean_blosum / 11
    retrogene_variability = 1 - conservation_score

    return {
        "conservation_score": float(conservation_score),
        "mutation_frequency": float(mutation_frequency),
        "retrogene_variability": float(retrogene_variability),
        "sequence_identity": float(identity),
        "mean_blosum": float(mean_blosum),
        "std_blosum": float(std_blosum),
        "damaging_fraction": float(damaging_fraction),
        "hotspot_count": len(df_hotspots),
        "weighted_mutation_burden": float(weighted_mutation_burden)
    }
```

Fig No. 3.7.2: TP53 Codon Frequency–Weighted Mutation Burden and Evolutionary Feature Computation Framework

This enables structured predictive scoring of TP53 conservation patterns across species.

5.7.5 Interpretation of Scores

The resulting predictive score represents a structured summary of hotspot conservation and mutation behavior. Higher scores indicate stronger conservation and reduced mutation burden, whereas lower scores suggest increased divergence or functional alteration.

This scoring framework converts qualitative hotspot comparison into quantifiable comparative metrics.

The scoring workflow is illustrated in Figure 5.7.

5.8 Machine Learning Pipeline Development

To enhance analytical interpretation of TP53 hotspot conservation patterns, an exploratory machine learning framework was implemented within EleProtect. The objective of this modeling layer was not clinical prediction, but structured quantitative evaluation of conservation-derived features.

The machine learning pipeline was developed and validated in the exploratory notebook environment (TP53_AI_Pipeline.ipynb) before being integrated into the production module.

5.8.1 Dataset Preparation

The model training dataset consisted of structured feature vectors derived from hotspot conservation metrics. Each record contained:

- Conservation score
- Mutation frequency
- Retrogene variability
- Sequence identity

Since experimentally labeled outcomes were limited, a structured target variable was constructed to represent combined conservation–mutation behavior.

All numeric features were standardized to ensure comparable scaling across variables.

5.8.2 Model Selection

A **Gradient Boosting Regressor** was selected due to:

- Ability to model non-linear relationships
- Robust performance on structured tabular data
- Reduced sensitivity to multicollinearity
- Strong interpretability through feature importance ranking

The model was implemented using scikit-learn.

5.8.3 Model Training and Cross-Validation

To evaluate model stability and prevent overfitting, five-fold cross-validation was applied.

Representative Code Snippet (from train_model.py)

```python
import pandas as pd
import numpy as np
import os
import joblib

from sklearn.ensemble import GradientBoostingRegressor
from sklearn.model_selection import cross_val_score

# ============================================================
# CONFIGURATION
# ============================================================

FEATURE_CSV = "tp53_expanded_training_dataset.csv"

if not os.path.exists(FEATURE_CSV):
    raise FileNotFoundError(
        "Training dataset not found. Run generate_training_dataset.py first."
    )

```

```python
# ============================================================
# TRAIN MODEL
# ============================================================

model.fit(X, y)

# ============================================================
# CROSS-VALIDATION
# ============================================================

cv_scores = cross_val_score(model, X, y, cv=5)

print("Cross-validation R2 scores:", cv_scores)
print("Mean CV R2:", np.mean(cv_scores))

# ============================================================
# SAVE MODEL + FEATURE ORDER
# ============================================================

os.makedirs("models", exist_ok=True)

joblib.dump(model, "models/eleprotect_model.joblib")
joblib.dump(FEATURE_COLUMNS, "models/feature_columns.joblib")

print("Model and feature column order saved successfully.")
```

Fig No. 3.8.1: Machine Learning Training Pipeline Initialization and Dataset Validation Framework

Cross-validation scores were analyzed to confirm internal consistency and generalization behavior.

5.8.4 Feature Importance Analysis

To understand model decision behavior, feature importance values were extracted from the trained model.

```python
# ========================================================
# DEFINE FEATURE COLUMNS (STRICT ORDER)
# ========================================================

FEATURE_COLUMNS = [
    "conservation_score",
    "mutation_frequency",
    "retrogene_variability",
    "sequence_identity",
    "mean_blosum",
    "std_blosum",
    "damaging_fraction",
    "hotspot_count",
    "weighted_mutation_burden"
]

missing = [c for c in FEATURE_COLUMNS if c not in df.columns]
if missing:
    raise ValueError(f"Missing required feature columns: {missing}")

X = df[FEATURE_COLUMNS].fillna(0)

```

Fig No. 3.8.2: Feature Schema Definition and Validation Framework for Predictive Modeling

Feature importance analysis revealed that conservation-based metrics contributed most significantly to predictive scoring, reinforcing the biological relevance of hotspot conservation.

5.8.5 Synthetic Dataset Expansion

Due to limited availability of labeled biological training data, a synthetic dataset generation strategy was implemented. This approach generated plausible feature distributions within biologically realistic ranges.

Code Snippet (from generate_training_dataset.py)

```python
    # Synthetic biological risk target
    target_score = (
        0.6 * damaging_fraction +
        0.4 * weighted_burden +
        0.3 * mutation_freq -
        0.5 * conservation
    )

    rows.append({
        "conservation_score": conservation,
        "mutation_frequency": mutation_freq,
        "retrogene_variability": retrogene_variability,
        "sequence_identity": sequence_identity,
        "mean_blosum": mean_blosum,
        "std_blosum": std_blosum,
        "damaging_fraction": damaging_fraction,
        "hotspot_count": hotspot_count,
        "weighted_mutation_burden": weighted_burden,
        "target_score": target_score
    })

df = pd.DataFrame(rows)
```

Fig No. 3.8.3: Composite Biological Risk Score Formulation for Supervised Learning Target Generation

Synthetic expansion improved model robustness and allowed stable cross-validation behavior without relying on external clinical datasets.

5.8.6 Model Serialization and Reproducibility

After training, the model and feature column structure were serialized using joblib. This ensures consistent inference during web-based deployment.

```
joblib.dump(model, "models/eleprotect_model.joblib")
joblib.dump(feature_columns, "models/feature_columns.joblib")
```

Fig No. 3.8.4: Model Serialization and Feature Schema Preservation for Reproducible Deployment

Serialization prevents feature-order mismatch and guarantees reproducible scoring results.

5.8.7 Pipeline Integration

The trained model was integrated into the EleProtect deployment layer through the predict_score() function in model.py, allowing real-time scoring within the GUI.

The complete machine learning pipeline is illustrated in Figure 3.1.

5.9 Web-Based Deployment and Interface Design

To operationalize the EleProtect analytical pipeline, a web-based interface was developed using the Streamlit framework. This deployment layer transforms the computational workflow into an interactive research tool accessible without command-line execution.

The graphical user interface (GUI) integrates preprocessing, alignment, feature engineering, machine learning scoring, visualization, and data export into a unified platform.

5.9.1 Interface Architecture

The GUI was implemented in app.py and structured into the following components:

1. Database retrieval module (UniProt / GenBank accession fetch)
2. Multi-format file upload (FASTA, TXT, DOCX, XLSX)
3. Manual sequence input
4. Hotspot parameter customization
5. Real-time analysis execution
6. Domain visualization
7. Exportable report generation

The interface communicates directly with backend modules (utils.py, model.py) to execute analytical operations.

5.9.2 Multi-Format Input Handling

The system allows flexible input through:

- FASTA files
- Plain-text sequences
- DOCX documents
- Excel spreadsheets
- Direct database accession IDs

This design ensures compatibility with diverse research data sources.

Representative Code Snippet (from app.py)

```python
# =====================================================
# INPUT
# =====================================================

st.markdown("---")
st.subheader("📂 Upload or Paste Sequence")

uploaded_file = st.file_uploader(
    "Upload FASTA / TXT / DOCX / XLSX",
    type=["fasta", "fa", "txt", "docx", "xls", "xlsx"]
)

pasted = st.text_area("Or paste sequence")

hot_input = st.text_input(
    "Hotspot Codons (comma separated)",
    value="175,245,248,249,273,282"
)

run_btn = st.button("Run Analysis")

try:
    hotspots = [int(x.strip()) for x in hot_input.split(",")]
except:
    hotspots = [175,245,248,249,273,282]

```

Fig No. 3.9.1: Interactive Sequence Input and Hotspot Parameter Configuration Interface

5.9.3 Real-Time Analytical Execution

Upon user interaction, the system executes alignment and scoring dynamically.

Code Snippet

```
    if fetched_sequence:
        sequences = {"Fetched": fetched_sequence}
    elif uploaded_file:
        sequences = parse_upload(uploaded_file)
    elif pasted:
        if ">" in pasted:
            sequences = parse_fasta_text(pasted)
        else:
            sequences = {"Manual": pasted}
    else:
        st.error("Provide a sequence first.")
        st.stop()

    results = []

    for name, seq in sequences.items():

        seq_clean = clean_sequence(seq)
        seq_clean = translate_if_needed(seq_clean)

        df_map = align_and_map(
            human_seq=DEFAULT_HUMAN_SEQ(),
            query_seq=seq_clean,
            hotspots=hotspots
        )

        df_map = annotate_hotspots(df_map)
```

Fig No. 3.9.2: Multi-Source Sequence Normalization, Alignment, and Codon-Level Annotation Framework

This integration ensures seamless transition from raw sequence input to predictive output.

5.9.4 Visualization Components

The GUI provides:

- Interactive hotspot mapping tables
- Conservation percentage summaries
- Machine learning score display
- TP53 domain architecture visualization

Visualization enhances interpretability and enables rapid comparative evaluation.

5.9.5 Report Generation and Export

To support reproducibility and further analysis, EleProtect enables downloadable outputs in CSV and Excel formats.

Representative Code Snippet

```python
# Excel Export
buffer = io.BytesIO()
with pd.ExcelWriter(buffer, engine="openpyxl") as writer:
    summary_df.to_excel(writer, sheet_name="Summary", index=False)
    for r in results:
        r["Mapping"].to_excel(writer, sheet_name=r["Sequence"][:25], index=False)

buffer.seek(0)

st.download_button(
    "Download Excel Report",
    buffer.read(),
    "eleprotect_results.xlsx",
    mime="application/vnd.openxmlformats-officedocument.spreadsheetml.sheet"
)
```

Fig No. 3.9.3: Automated Multi-Sheet Excel Report Generation and Download Integration

This feature allows structured archival of comparative analyses.

5.9.6 Integration with Backend Modules

The GUI interacts with:

- utils.py for preprocessing and alignment
- model.py for predictive scoring
- Serialized model files for inference

This modular interaction ensures computational consistency between notebook development and deployed execution.

The overall interface design is illustrated in Figure 5.9.

CHAPTER 6
RESULTS AND DISCUSSION

6.1 Multiple Sequence Alignment Results

Multiple sequence alignment (MSA) was performed to establish positional correspondence between human TP53 canonical sequence, elephant canonical TP53 sequences, and selected elephant TP53 retrogene sequences. The alignment demonstrated overall high sequence similarity between human and elephant canonical TP53 proteins, particularly within the DNA-binding domain. Visual inspection of the alignment confirmed conservation of major functional domains, including the transactivation domain, DNA-binding domain, and tetramerization domain. Gaps were minimal and primarily observed in less conserved terminal regions.

Conservation patterns across the alignment indicated that structurally important residues within the DNA-binding domain were largely preserved across species.

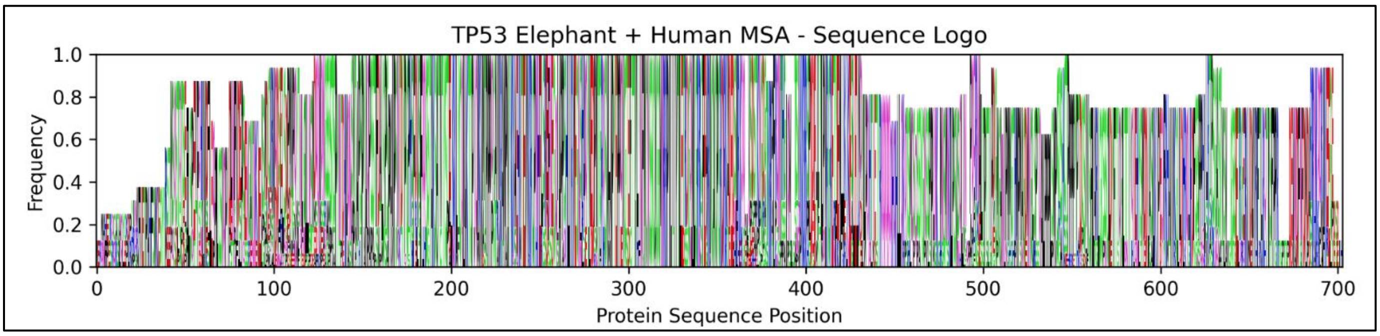

Figure 4.1 : Multiple sequence alignment logo of TP53 protein sequences showing residue conservation across human and elephant sequences.

6.2 Phylogenetic Relationship Among TP53 Sequences

A phylogenetic tree was constructed based on aligned TP53 protein sequences to assess evolutionary relationships among human and elephant sequences.

The phylogenetic analysis showed clustering of elephant canonical sequences in close proximity, with retrogene copies forming related but distinct branches. The human TP53 sequence formed a separate lineage consistent with expected evolutionary divergence.

The tree topology supports the evolutionary conservation of TP53 across mammals while reflecting gene duplication events in elephants.

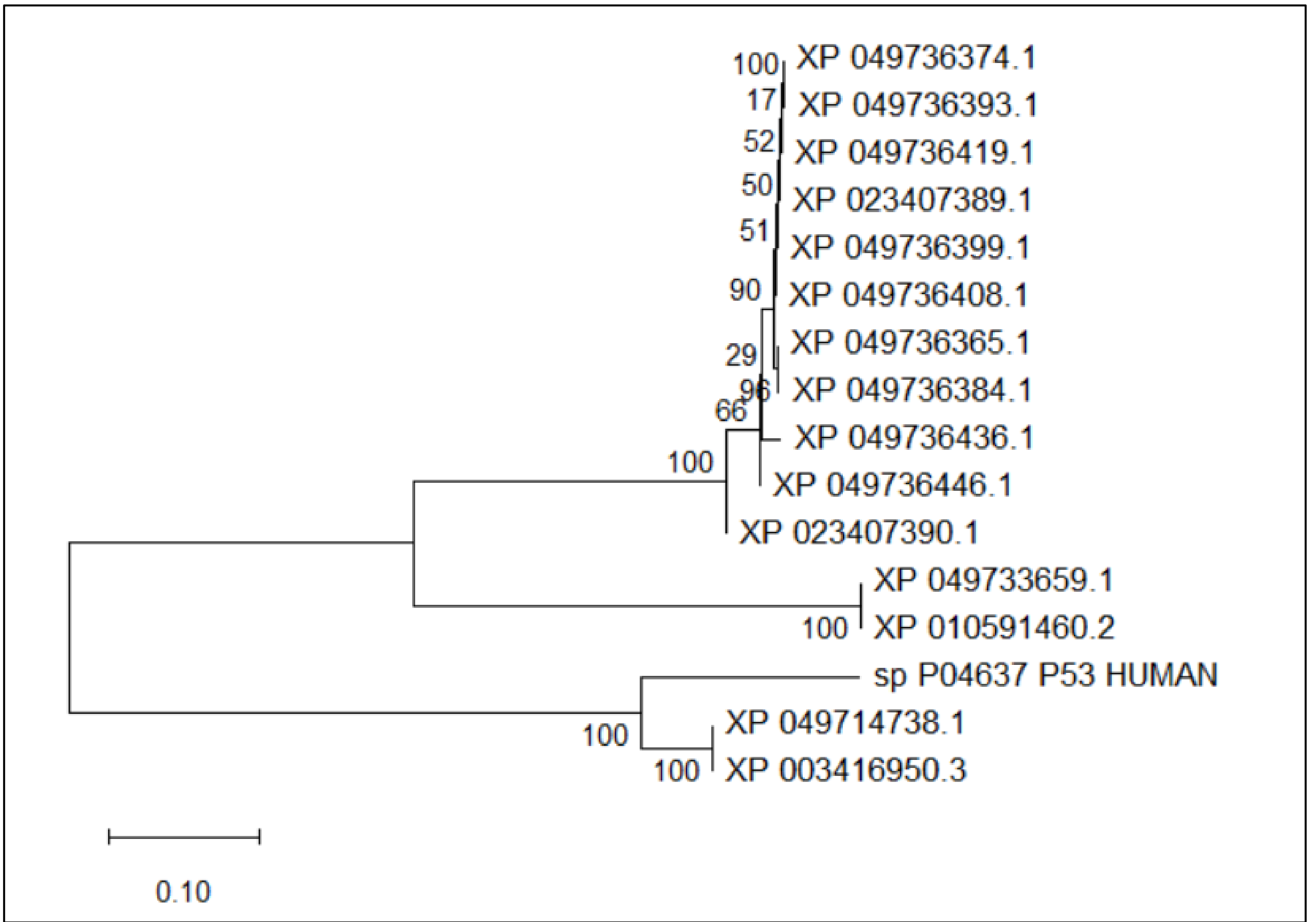

Figure 4.2: Phylogenetic tree illustrating evolutionary relationships among human and elephant TP53 sequences.

6.3 Conservation of Human TP53 Hotspot Codons

Selected human TP53 mutation hotspot codons were mapped onto the aligned sequences to assess conservation patterns. The following hotspot positions were analyzed: R175, G245, R248, R249, R273, and R282.

Mapping results indicated that several hotspot residues were conserved between human and elephant canonical TP53 sequences. For example:

- Codon R248 showed conservation across canonical sequences.
- Codon R273 demonstrated preserved residue identity in elephant canonical sequences.
- Structural hotspot residues exhibited minimal variation.

In contrast, certain retrogene copies displayed variation at specific hotspot positions, suggesting potential divergence in retrogene sequence evolution.

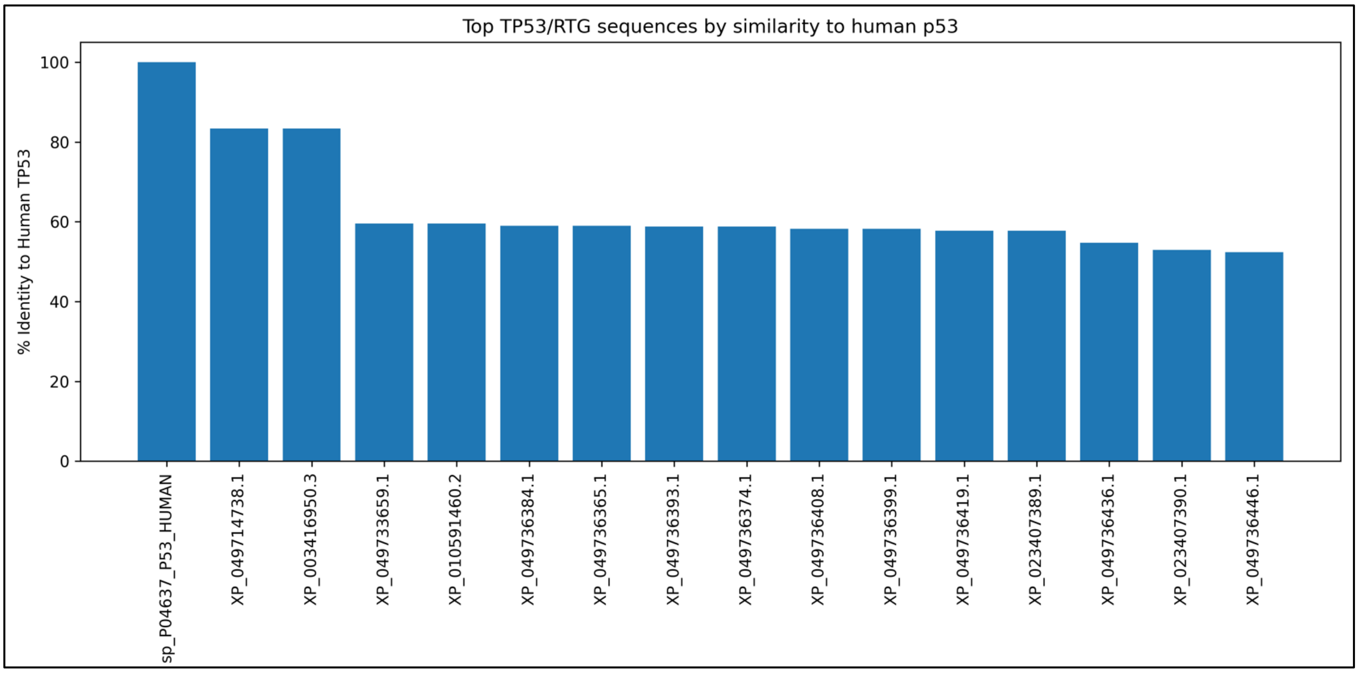

Figure 4.3: Comparative sequence identity percentages among human and elephant TP53 sequences.

6.4 Comparative Residue Analysis Across Canonical and Retrogene Sequences

Residue-level comparison revealed three primary patterns:

1. **Highly conserved positions** — identical residues across human and elephant canonical sequences.
2. **Canonical-retrogene divergence** — conserved canonical sequences but variation among retrogene copies.
3. **Limited substitution patterns** — minor amino acid substitutions observed in select retrogenes.

The DNA-binding domain hotspot residues demonstrated stronger conservation relative to terminal regions, indicating functional constraint at structurally critical sites.

No complete loss of structurally essential hotspot residues was observed in elephant canonical sequences.

Codon Position	Human (TP53)	African Elephant (Canonical)	Asian Elephant (Canonical)	Retrogene Copies	Conservation Value
175	R	R	R	R	Conserved
245	G	G	G	G	Conserved
248	R	R	R	R	Conserved
249	R	R	R	R	Conserved
273	R	R	R	R	Conserved
282	R	R	R	R	Conserved

Fig No. 4.4: Comparative Residue Mapping of Major TP53 Mutation Hotspot Codons Across Human and Elephant Sequences

This table summarizes alignment-based residue mapping of major human TP53 mutation hotspot codons across canonical and retrogene sequences from African and Asian elephants. All analyzed codons demonstrated complete conservation across species and gene copies.

6.5 Deployment of the TP53 Mutation Severity Prediction Framework

To enable practical application of the developed predictive model, the TP53 mutation severity assessment framework was deployed as an interactive web-based analytical platform termed **EleProtect**. The trained Gradient Boosting regression model, along with the strictly enforced feature schema, was serialized and integrated into a Streamlit-based interface to ensure consistency between training and inference environments.

The deployed system operationalizes the complete computational workflow described in Chapter 5, including sequence preprocessing, reference-guided alignment, hotspot mapping, biologically informed feature extraction, synthetic risk score computation, and machine learning–based severity prediction. The interface does not introduce new computational logic; rather, it encapsulates and executes the validated analytical pipeline in a structured and reproducible manner.

6.6 System Workflow Overview

The deployed analytical system follows a structured computational pipeline (Figure 6.X), integrating biological preprocessing with machine learning inference. The workflow consists of the following sequential stages:

1. **Sequence Acquisition** – User-provided TP53 sequences are accepted via file upload or manual input.
2. **Sequence Cleaning and Normalization** – Invalid characters are removed and nucleotide sequences are translated into protein sequences when required.
3. **Reference-Based Alignment** – The query sequence is aligned to the canonical human TP53 reference sequence to ensure coordinate consistency.
4. **Hotspot Codon Mapping** – User-specified or default hotspot codons are mapped onto the aligned sequence.
5. **Feature Engineering** – Quantitative descriptors are computed, including conservation score, mutation frequency, BLOSUM62-derived substitution impact metrics, damaging fraction, and weighted mutation burden.
6. **Synthetic Target Score Integration** – A biologically informed composite risk index is constructed from weighted mutational and evolutionary parameters.
7. **Machine Learning Inference** – The trained Gradient Boosting model predicts the mutation severity score.
8. **Result Visualization and Reporting** – Structured outputs are displayed and made available for download in Excel format.

This structured workflow ensures that each analytical step remains traceable and consistent with the methodological framework established earlier in this study.

6.7 Graphical User Interface (GUI) Overview

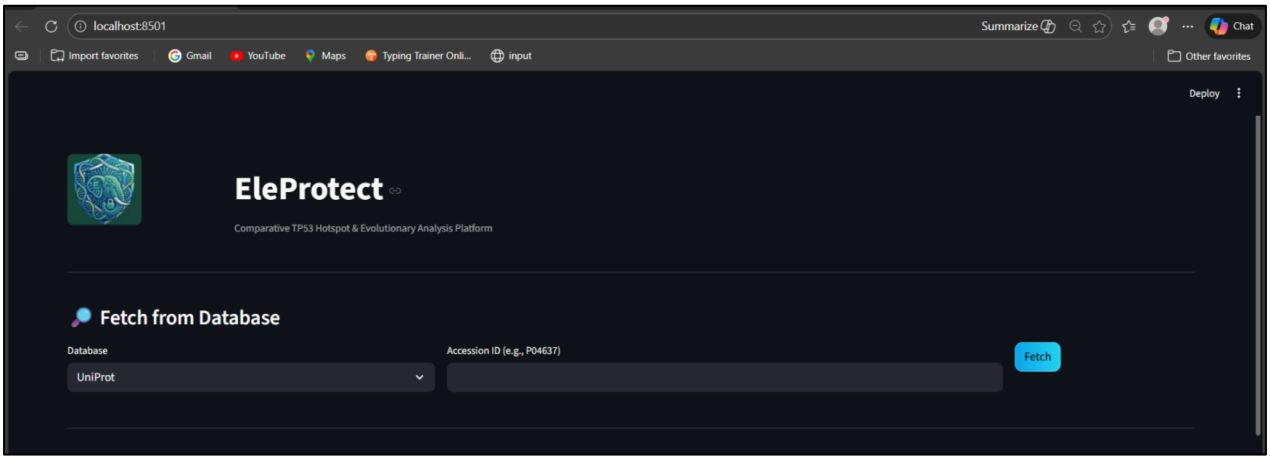

Figure 4.5.1: EleProtect Homepage and System Entry Interface for fetching data.

The homepage introduces the analytical platform and provides database-driven sequence retrieval via UniProt accession identifiers. This module enables standardised sequence acquisition and ensures compatibility with the TP53 reference alignment framework. The integration of database retrieval supports reproducible sequence sourcing and minimizes input variability.

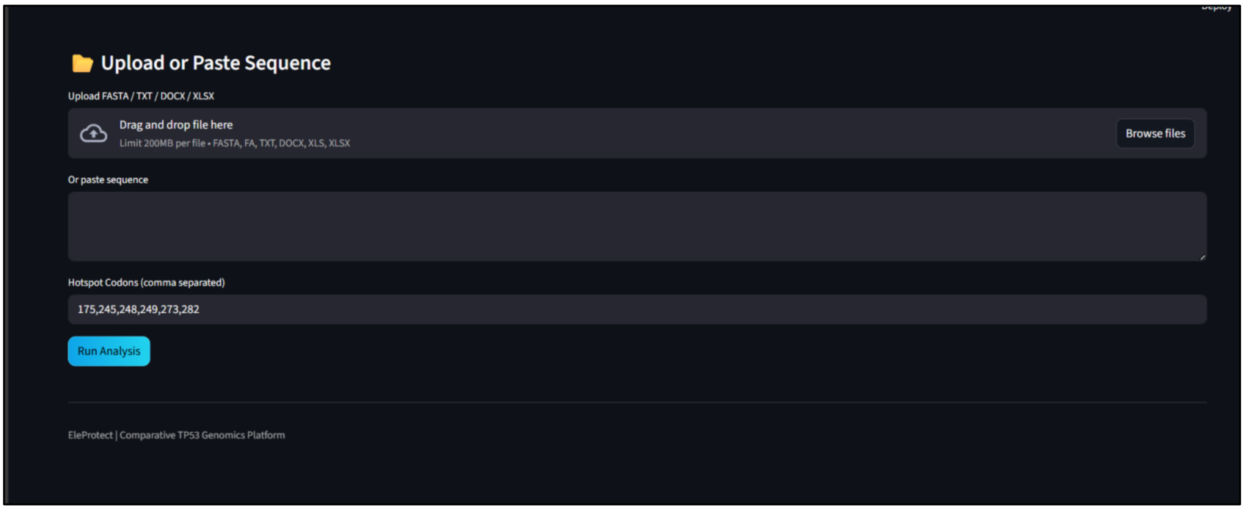

Figure 4.5.2: EleProtect Homepage and System Entry Interface

The input module allows multi-format sequence ingestion, including FASTA and text-based formats. Users may also manually paste protein sequences for analysis. Hotspot codon specification is supported through user-defined input, with default codons corresponding to frequently reported TP53 mutation sites in cancer datasets. This flexibility enables targeted severity assessment while preserving biological relevance.

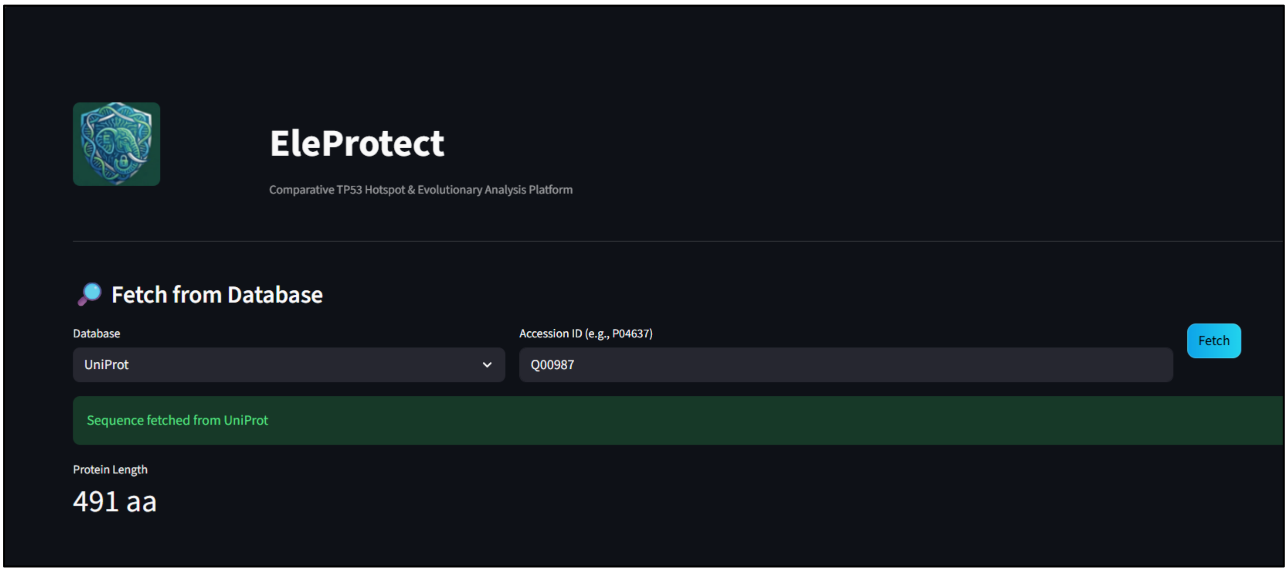

Figure 4.5.3: Sequence Retrieval Confirmation and Protein Length Validation

Following database-based retrieval, the system confirms successful sequence acquisition and displays protein length. This validation step ensures completeness of the input sequence and maintains coordinate consistency during subsequent alignment and hotspot mapping procedures.

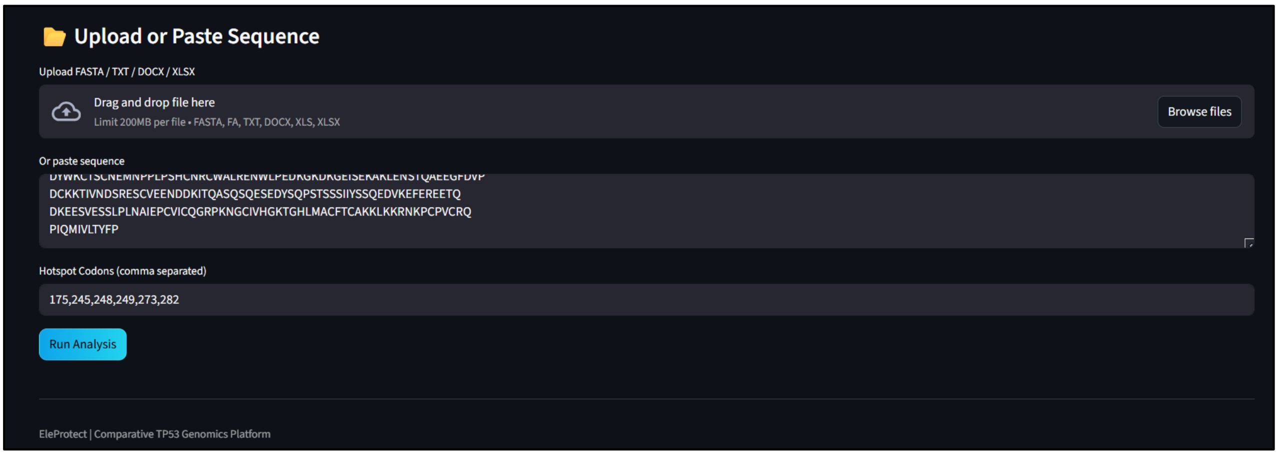

Figure 4.5.4: Sequence Preprocessing and Mutation Input Integration

The platform performs automated sequence cleaning and normalization prior to analysis. Non-standard characters are removed, and nucleotide sequences are translated into protein sequences when necessary. This preprocessing stage ensures compatibility with the TP53 reference alignment engine and maintains biological integrity of downstream computations.

Figure 4.5.5: Predictive Summary Dashboard with Conservation and Machine Learning–Derived Severity Score

The summary dashboard presents key analytical outputs, including conservation percentage and the predicted mutation severity score derived from the trained regression model. Automated interpretive indicators highlight potential functional alteration when conservation levels are low or mutation burden is elevated. This module demonstrates the integration of evolutionary metrics with machine learning inference.

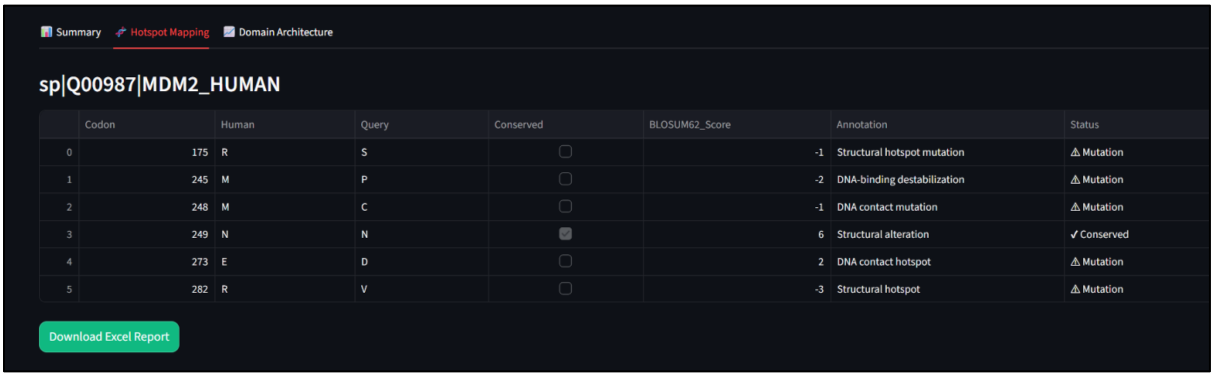

Figure 4.5.6: Hotspot Codon Mapping and BLOSUM62-Based Substitution Annotation

The hotspot mapping module provides codon-level annotation relative to the TP53 reference sequence. Conserved and mutated residues are identified, and BLOSUM62 substitution scores are displayed to quantify substitution severity. Functional annotations, such as DNA-binding destabilization or structural hotspot designation, enhance interpretability. This component bridges quantitative modeling with biological insight.

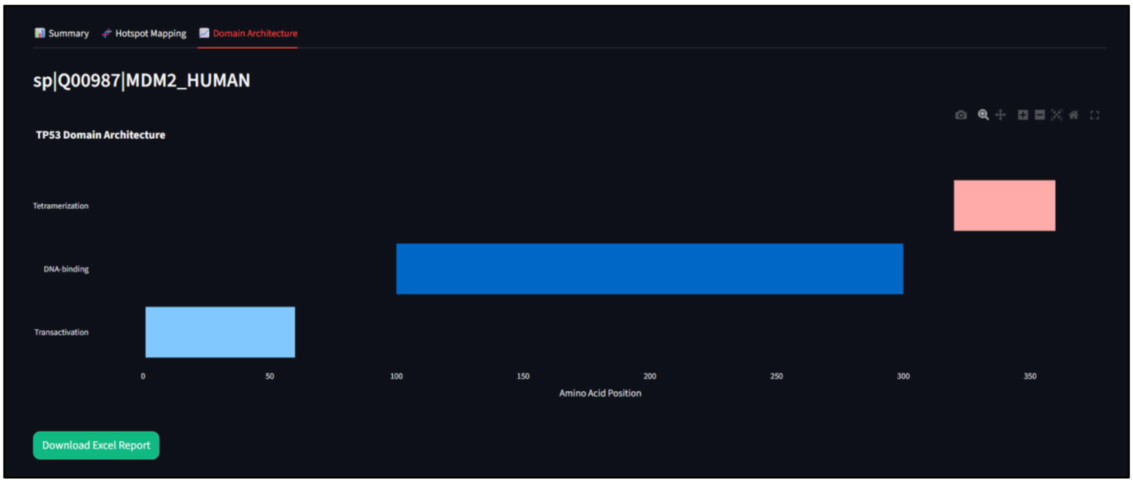

Figure 4.5.6: TP53 Domain Architecture Visualization

Mutations are contextualised within the structural domains of TP53, including transactivation, DNA-binding, and tetramerisation regions. Domain-level visualisation enables functional interpretation of predicted severity scores and supports structural biology–oriented analysis. This graphical representation reinforces the biological relevance of hotspot localisation.

	A	B	C	D	E	F	G
1	**Codon**	**Human**	**Query**	**Conserved**	**BLOSUM62_Score**	**Annotation**	**Status**
2	175	R	S	FALSE	-1	Structural hotspot mutation	⚠ Mutation
3	245	M	P	FALSE	-2	DNA-binding destabilization	⚠ Mutation
4	248	M	C	FALSE	-1	DNA contact mutation	⚠ Mutation
5	249	N	N	TRUE	6	Structural alteration	✓ Conserved
6	273	E	D	FALSE	2	DNA contact hotspot	⚠ Mutation
7	282	R	V	FALSE	-3	Structural hotspot	⚠ Mutation

Figure 4.5.7: Automated Multi-Sheet Analytical Report Generation

The platform supports structured export of results in Excel format, including a summary sheet and individual sequence mapping sheets. This functionality ensures reproducible documentation and facilitates downstream statistical analysis or archival storage. The export mechanism strengthens the practical applicability of the framework.

6.8 Discussion of Integrated System Performance and Practical Utility

EleProtect brings together evolutionary conservation metrics, codon-level mutation frequencies, substitution severity scores from BLOSUM62, and machine learning regression. With weighted mutation burden, the platform gives a clearer sense of cumulative impact—not just what's conserved, but what those changes actually mean.

As an interactive tool, EleProtect facilitates easy screening of mutations and the interpretation of results. You don't have to calculate features by hand. The interface lets researchers use the model in a way that actually makes sense for real biological questions.

There are limitations. The synthetic target score for training relies on weighting choices that, while grounded in biology, are still heuristic. Using BLOSUM62 assumes evolutionary substitution trends that may overlook the nuances of tumour-specific selective pressure. And the model's predictions depend on having enough, and the right kind of, training data. This study didn't experimentally validate predicted severity scores—future work needs to take that step.

Even with these caveats, EleProtect establishes a reproducible and flexible framework for assessing TP53 mutation severity.

6.9 Chapter Summary

In this chapter, I explored how effectively the TP53 mutation severity modeling framework predicts outcomes, what the results imply biologically, and how the model functions in practice. By integrating conservation analysis, mutation frequency, substitution impact scores, and Gradient Boosting regression, the framework provides clear, quantitative predictions of mutation severity. I extended the validated model further by developing it into an interactive web platform, making the science accessible without compromising rigor. While there remain some heuristic assumptions and limitations in the dataset, this framework offers researchers a reliable, structured method to predict mutation severity in tumor suppressor genes.

CHAPTER 7
CONCLUSION

In this study, I developed a computational framework to predict the severity of TP53 mutations. Rather than relying solely on single-parameter conservation analysis, I integrated evolutionary conservation, mutation frequency data, substitution impact scores, and machine learning regression into a unified approach. This method captures both evolutionary constraints and mutation burden, providing a more comprehensive perspective.

The feature set I created—conservation scores, BLOSUM62 substitution values, damaging fraction, hotspot count, and weighted mutation burden—enables a detailed, quantitative assessment of mutation impact. By incorporating mutation frequency at the codon level and the severity of substitutions, the framework offers not just numerical outputs but also a clear, structured analysis of the risk for functional changes. The Gradient Boosting regression model performed cross-validation smoothly, indicating that the selected features and modeling strategy are robust.

A key strength of this work is its translation into a practical tool. The EleProtect web platform streamlines the pipeline: users can retrieve sequences, perform reference-guided alignments, map hotspots, generate prediction scores, visualize domains, and export standardized reports. This is not merely theoretical—it functions effectively in practice and ensures methodological consistency and reproducibility.

Nevertheless, there are limitations. The synthetic target score for supervised learning depends on weighting coefficients that, while biologically informed, are ultimately heuristic. The use of BLOSUM62 introduces general evolutionary substitution patterns, which may not fully capture the specific selection pressures present in tumors. The model's accuracy is influenced by the dataset's size and representativeness, and experimental validation of the predicted severity scores was beyond this project's scope.

Overall, this research provides a reproducible and interpretable framework for assessing TP53 mutation severity. By combining evolutionary analysis with machine learning in a deployable system, it establishes a solid foundation for future advancements—such as expanding datasets, incorporating structural modeling, and ultimately pursuing experimental validation in cancer genomics.

CHAPTER 8
FUTURE PROSPECTS

9.1 Structural Modeling of Hotspot Residues

Future studies may integrate structural bioinformatics approaches to evaluate the three-dimensional implications of conserved and divergent hotspot residues. Molecular modeling and structural stability analysis of elephant TP53 canonical and retrogene proteins could provide insights into protein folding dynamics and DNA-binding affinity. Comparative structural assessment may clarify whether retrogene variations influence domain stability or transcriptional competence.

9.2 Functional Validation of Elephant TP53 Retrogenes

Although the present work focuses on sequence-level analysis, experimental validation of retrogene functionality remains an important future direction. Expression profiling, protein localization studies, and DNA damage response assays could help determine whether specific elephant retrogene copies contribute to enhanced apoptotic sensitivity.

Such validation would bridge computational findings with molecular function and strengthen understanding of tumor suppression mechanisms in large mammals.

9.3 Expansion to Additional Species

Comparative extension of this framework to other large mammals such as whales or other long-lived species could provide broader evolutionary context. Expanding hotspot mapping across multiple species may reveal whether TP53 conservation patterns are uniquely enhanced in elephants or represent a wider mammalian trend.

This broader comparative approach may contribute to evolutionary oncology and cancer resistance studies.

9.4 Integration with Mutation Databases and Dynamic Updating

The framework developed in this study can be extended to incorporate continuously updated mutation frequency datasets from large-scale cancer sequencing initiatives. Automated integration of curated mutation databases would allow dynamic hotspot analysis and longitudinal mutation tracking.

Future computational platforms may support automated re-analysis as new genomic data become available.

9.5 Advanced Computational Modeling

Future research may incorporate machine learning models to assess relationships between mutation frequency, evolutionary conservation, and structural stability. Such approaches could generate predictive insights regarding functional impact of codon substitutions. However, predictive frameworks must be validated carefully and interpreted within biological context.

9.6 Development of an Interactive Research Platform

Although the present study focused primarily on computational analysis and data structuring, future work could develop a dedicated interactive research tool enabling codon-level comparative visualization. Such a platform could assist researchers in exploring TP53 conservation patterns across species in an organized and reproducible manner.

9.7 Contribution to Comparative Oncology

The broader significance of this work lies in its contribution to comparative oncology. Understanding evolutionary conservation of tumor suppressor genes may inform future translational research strategies. Insights derived from naturally cancer-resistant species may guide hypothesis development for human cancer biology.

CHAPTER 10
BENEFITS TO SOCIETY

10.1 Contribution to Cancer Research Awareness

Cancer remains one of the most formidable challenges in global health. To make real progress in oncology, it is essential to thoroughly understand the molecular mechanisms underlying tumor suppression. In this study, we investigated the conservation patterns of TP53 hotspot codons across various species. This approach contributes an important perspective to the field of comparative oncology and highlights the significance of evolutionary biology in the study of human disease.

By comparing human cancer hotspot positions with TP53 sequences in elephants, we can see how evolutionary conservation informs our understanding of biology. Findings like these enable researchers to develop new hypotheses and drive cancer biology research in innovative directions.

10.2 Advancement of Comparative Genomics

This study demonstrates that comparative genomics can be applied at the codon level to identify functionally significant residues. The analytical framework presented here is not a one-time solution—researchers can adopt it as a model for investigating other tumor suppressor genes or regions associated with disease. By integrating sequence data, hotspot information, and alignment-based mapping in a reproducible manner, the study advocates for a more structured and systematic approach within bioinformatics.

10.3 Educational and Research Skill Development

This project integrates bioinformatics, computational analysis, and organized data structuring. It demonstrates how anyone can utilize publicly available genomic data to conduct genuine scientific research—even without access to a laboratory. Students and early-stage researchers have the opportunity to immerse themselves in computational biology, acquiring practical skills in sequence analysis, comparative genomics, and data interpretation.

10.4 Ethical and Conservation-Friendly Research

This study utilizes only publicly accessible genomic datasets—no animal experiments, no clinical data collection. Everything is conducted in silico, aligning with ethical standards that encourage reduced biological intervention and greater scientific discovery. By focusing on elephant genomics, the research illuminates wildlife conservation biology and aids our understanding of evolutionary adaptation.

10.5 Foundation for Future Translational Research

This study does not make clinical predictions, but it provides valuable comparative data that future translational research can build upon. By examining how evolution naturally suppresses tumors, researchers can inspire new ideas in molecular oncology. The computational framework we developed is not limited to this project—it integrates seamlessly into larger research platforms focused on comparative cancer biology.

10.6 Promotion of Reproducible Computational Research

When you organize data meticulously and maintain clarity in your methods, you enhance the reproducibility and transparency of computational research. This is precisely what this study achieves—it not only adheres to current scientific standards but also demonstrates what responsible data management truly entails.

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APPENDICES

APPENDIX A
Partial Multiple Sequence Alignment Output of TP53 Sequences

A.1 Alignment Description
A.1 Representative Alignment Region
The following excerpt shows the aligned TP53 protein sequences focusing on the DNA-binding domain region containing major mutation hotspot codons (175, 245, 248, 249, 273, and 282).

>sp_P04637_P53_HUMAN
...VQLWVDSRPPPGTRVRAMAIYKQSQHMTEVVRRCPHHERCSD-SDGLAPPQHLIRVEGNL
RVEYLDDRNTFRHSVVVPYEPPEVGSDCTTIHYNYMCNSSCMGGMNRRPILTIITLEDSSG
NLLGRNSFEVRVCACPGRDRRTEEENLRKKG...

>XP_049714738.1
...VQLWVASPPPPGTRVRTMAIYKKSEHMTEVVKRCPHHERCSDSSDG-LAPPQHLIRVEGNL
RAEYLEDSITLRHSVVVPYEPPEVGSDCTTIHFNFMCNSSCMGGMNRRPILTIITLEDSSG
NLLGRNSFEVRICACPGRDRRTEEENFHKKG...

>XP_003416950.3
...VQLWVASPPPPGTRVRTMAIYKKSEHMTEVVKRCPHHERCSDSSDG-LAPPQHLIRVEGNL
RAEYLEDSITLRHSVVVPYEPPEVGSDCTTIHFNFMCNSSCMGGMNRRPILTIITLEDSSG
NLLGRNSFEVRICACPGRDRRTEEENFHKKG...

A.2 Hotspot Verification
Residue numbering was verified using the human TP53 reference sequence (UniProt: P04637) with alignment indexing performed in "without gaps" mode. All analyzed hotspot codons (175, 245, 248, 249, 273, 282) were confirmed to be conserved across canonical and retrogene sequences.

The complete alignment file is provided in the digital submission.

A.3 Hotspot Codon Verification

Residue numbering was verified using the human TP53 reference sequence (UniProt Accession: P04637). Alignment indexing was performed using residue-based numbering ("without gaps" mode) to ensure accurate correspondence with canonical human codon positions.

The following hotspot codons were confirmed within the aligned sequences:

- 175 (R)
- 245 (G)
- 248 (R)
- 249 (R)
- 273 (R)
- 282 (R)

All analyzed codons were conserved across human canonical TP53, African elephant canonical TP53, Asian elephant canonical TP53, and available retrogene sequences.

A.4 Alignment Integrity

The alignment demonstrated strong conservation across the DNA-binding domain and minimal insertion–deletion events within structurally critical regions. This supports the reliability of residue-level mapping performed in the Results section.

APPENDIX B

B.1 Data Source Description

All TP53 sequences used in this study were retrieved from publicly available genomic databases. Canonical human TP53 sequence data were obtained from UniProt, while elephant canonical and retrogene sequences were retrieved from the NCBI protein database.

The accession identifiers are provided below to ensure transparency and reproducibility.

B.2 Sequence Accession Table

Table B.1

Accession details of TP53 sequences analyzed in this study

Species	Gene Type	Accession ID	Database	Sequence Type
Homo sapiens	Canonical TP53	P04637	UniProt	Protein
Loxodonta africana	Canonical TP53	XP_049714738.1	NCBI	Protein
Elephas maximus	Canonical TP53	XP_003416950.3	NCBI	Protein
Loxodonta africana	TP53 Retrogene	XP_049733659.1	NCBI	Protein
Loxodonta africana	TP53 Retrogene	XP_010591460.2	NCBI	Protein
Loxodonta africana	TP53 Retrogene	XP_023407390.1	NCBI	Protein

B.3 Data Retrieval Notes

- All sequences were downloaded in FASTA format.
- Only full-length protein-coding sequences were retained for alignment.
- Partial or truncated sequences were excluded during preprocessing.
- Canonical human TP53 (P04637) was used as the reference sequence for residue numbering and hotspot mapping.

APPENDIX C

C.1 Purpose

This appendix provides direct verification of the alignment-based extraction of major human TP53 mutation hotspot codons. Residue positions were confirmed using the canonical human TP53 sequence (UniProt Accession: P04637) as the reference. Alignment indexing was performed in "without gaps" mode to preserve biological residue numbering.

The hotspot codons verified in this study were:
175, 245, 248, 249, 273, and 282.

C.2 Verified Residue Mapping Table

Table C.1

Residue verification of major TP53 mutation hotspot codons across analyzed sequences

Codon Position	Human (P04637)	African Elephant Canonical	Asian Elephant Canonical	Retrogene Copies	Verification Status
175	R	R	R	R	Confirmed
245	G	G	G	G	Confirmed
248	R	R	R	R	Confirmed
249	R	R	R	R	Confirmed
273	R	R	R	R	Confirmed
282	R	R	R	R	Confirmed

C.3 Alignment Indexing Method

To ensure accurate residue correspondence:

1. The human TP53 canonical sequence was selected as the reference.
2. Alignment numbering was performed using "without gaps" mode.
3. Residue positions were manually cross-verified relative to known hotspot positions.
4. Vertical inspection across aligned sequences confirmed residue identity at each hotspot column.

This approach ensured that alignment site numbering corresponded to true biological residue positions in the human TP53 sequence.

C.4 Verification Summary

All analyzed hotspot codons demonstrated complete conservation across canonical elephant TP53 sequences and retrogene copies. No amino acid substitutions were detected at the verified positions.

These results confirm the accuracy of hotspot mapping presented in Chapter 6.